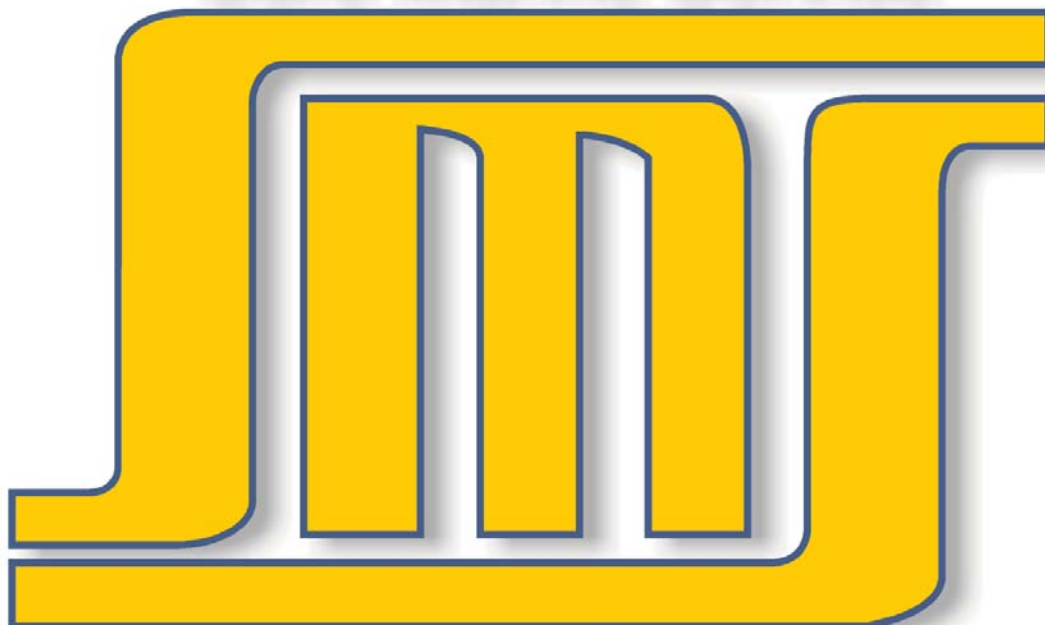


Sure Marine Service ®



[www.suremarine.com](http://www.suremarine.com)

# DBW-2010 Installation Guide

With SureWire™ Control



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June 9th 2008

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Visit our showroom to see and feel these heaters in operation.



The text, drawings and other artwork are not to scale and are simply intended to be guides to aid in your selection and installation process. They do not replace manufacturer or owners manuals. We will provide operation restrictions for the unit, but certain areas have legal restrictions that must be heeded as well.

Ensure all parts of the heater's installation are within approved standards and workmanship.

Non-approved components may not meet operational requirements of this heating system and may result in failure of that component or of the entire system.

The content of this document is furnished for informational use only. It is subject to change without notice. Sure Marine Service assumes no responsibility or liability for any errors or inaccuracies that may appear in the content.

The information in this packet should be read and followed. Any mechanical problem due to failure to follow these installation requirements will not be covered under warranty.

This manual in no way removes you from any legal obligation that may be present in your area, nor does it make SMS liable for any installation done by our customers. Reference sources that should also be consulted (though much of this manual has done this already) are ABYC standards, NFPA codes and USCG standards.

# Marine Installation Guide

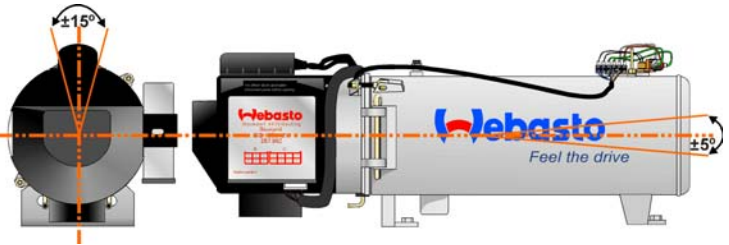
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## Properly Positioning Heating System Components

The first step in installing a Webasto boiler in your boat is to map out the location of the system components. Each system component serves a specific purpose for the proper operation of the system. Proper location of a component is an important factor in the continued performance of your heating system. Improper placement of a component may lead to the component's failure and ultimately a system failure that may result in heavy damage to the boiler itself.

### Boiler Location

Preferably the heater would be mounted facing fore and aft with the exhaust out the bottom. If this is not possible, the heater may be mounted within the limits shown in the diagram.



The boiler itself is subject to more restrictions on its placement than the rest of the system. It must meet the restrictions of its exhaust system, electrical system and the water system. Read this entire manual before beginning work to ensure all restrictions are met by your choice of location.

Because of the operation noise of the unit and system restrictions that may affect location choices, the engine room is the most common area chosen for installation. The engine room in most boats has a large area that can accommodate the size of the heater and allow for room for other system components such as the system pump and the hot water tank.



**If the boat has gas powered engines, the unit may not be placed inside the gas fuel tank area, engine room or below the carburetor line of the boat. This is due to spark arrest restrictions for gas powered boats.**

**If the heater is installed in the bilge or engine room, 100% fresh air must be supplied for combustion air per ABYC standards.**

ABYC standards as well as fire code do not permit non-spark arrested units inside gas engine rooms. The boiler is not spark-arrested. If your boat is gas-powered, an alternate location for your heater must be found.

If the engine room is not a possible location, a large lazarette or void space may be used, keeping heater system restrictions in mind. The boiler generates substantial amounts of heat and combustion uses a large volume of air. Proper ventilation is critical.

The aft area of both power and sail boats are used frequently with good results. Mounting the boiler in the aft area simplifies routing the exhaust to the transom which is usually a relatively dry location away from waves and spray. By routing the exhaust aft instead of out the side you can avoid blowing exhaust onto the dock or onto rafted boats.



**Do not forget to leave room for the powerhead to swing open for service.**

There must be 12" to the side of the heater to allow the powerhead to swing open and 12" in front of the powerhead to allow for combustion tube removal.

### Exhaust Routing



The exhaust gasses must be piped overboard. It is a high-temperature dry exhaust that requires insulation. Remember that if your thru-hull does not exit out of the transom, your exhaust becomes a hazard to boats that may raft up to you. The temperatures of the exhaust system and gas can exceed 800° F.

The system consists of rigid exhaust tube bent to necessary angles to install in your boat, an exhaust thru-hull, exhaust clamps, exhaust sealant and insulation blanket. You will want to situate the boiler so that you can properly run the exhaust pipe maintaining a minimum 1" clearance from the outside of the exhaust insulation to wood or any other flammable material.

## Water System

**For proper operation the system should hold at least 10 gallons of water/antifreeze coolant.**



It is important for the proper operation of your heating system that the expansion tank is located above all components in your water system. This is to maintain head pressure on the system and to prevent air from interfering with the flow of water in the system. If any component is located above the expansion tank, air may be trapped in this component and restrict the flow of water through the system. Reduced flow can lead to poor performance of the heating system and potential damage to the water pump.

Your layout should have an unrestricted flow of water from the expansion tank to the system pump. Components that may restrict the flow of water such as fan heaters or hot water tanks should not be placed between the expansion tank and the pump. Components that do not restrict the flow of water such as engine heat exchangers may be placed in this line.

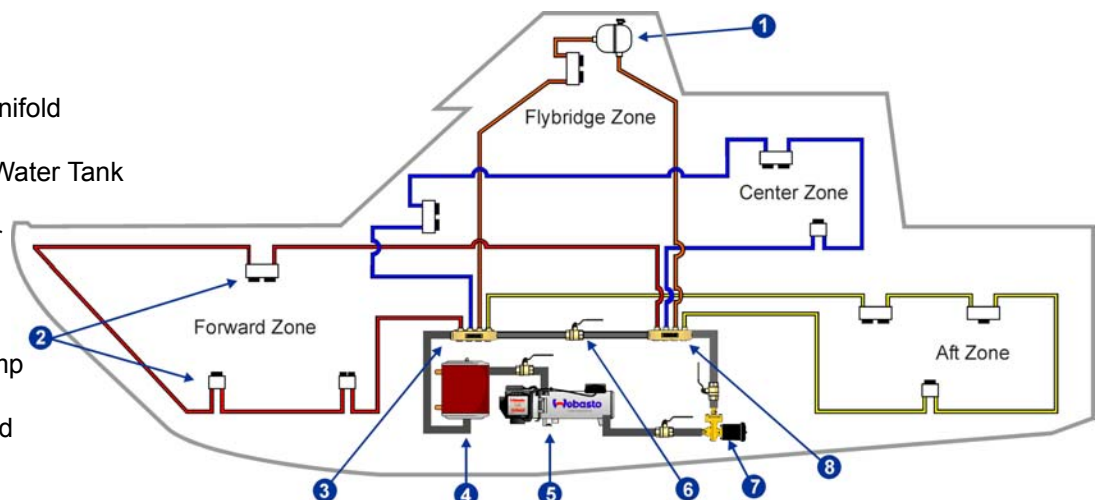
A direct line should also exist between the pump and the heater. The pump should have no components except free-flow shut-off valves between itself and the boiler.

Fan heaters can be installed anywhere in the system, provided they are installed below the expansion tank and not in the line from the expansion tank to the pump or boiler.

Like the fan heaters, heat exchangers must be below the expansion tank. However, because they are free-flowing they may be installed in between the expansion tank, system pump and heater.

A common marine layout might have a forward and an aft loop. In boats that have a flybridge, a loop can be installed to run flybridge heat and to install the expansion tank.

- 1 Expansion Tank (must be highest point of system)
- 2 Fan Heater(s)
- 3 Distribution Manifold
- 4 Domestic Hot Water Tank
- 5 Webasto Boiler
- 6 Summer Valve
- 7 Circulation Pump
- 8 Return Manifold



*Marine Layout Example*

## Expansion Tank



The expansion tank must be located at the highest point of the heating system and connected as directly as possible to the inlet of the circulation pump.

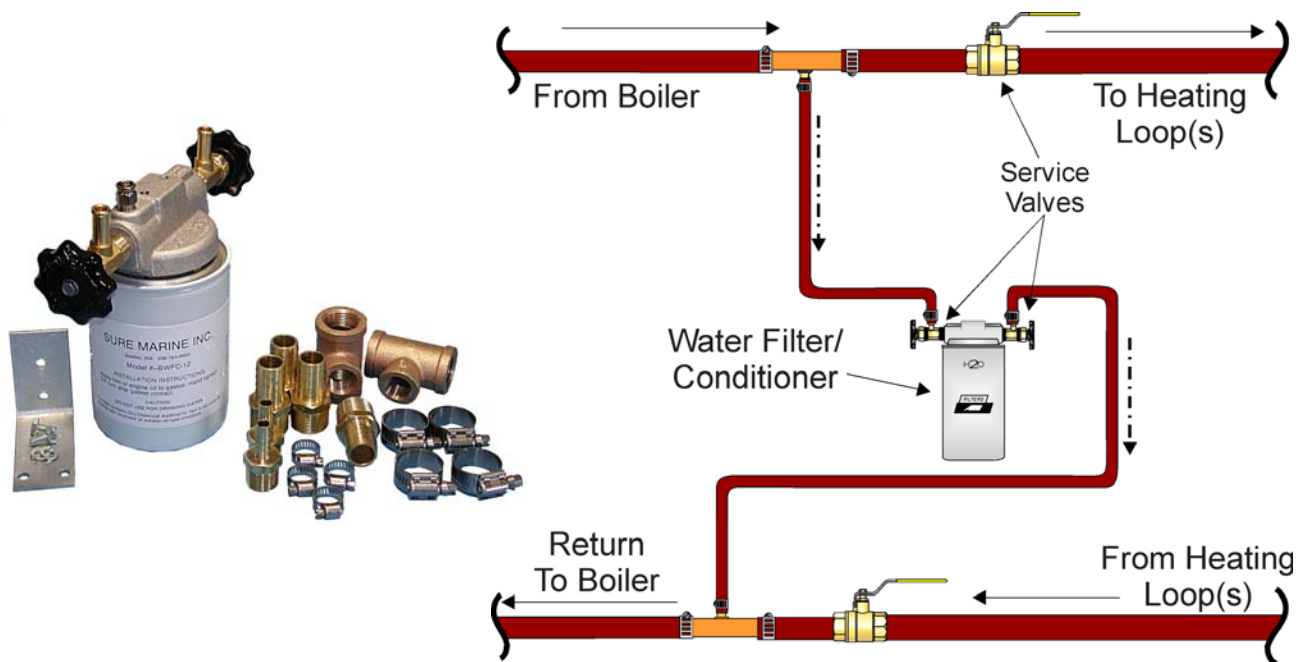
This kind of system must allow for expansion and contraction of system components as they heat up and cool down. The expansion tank accommodates this by allowing the water level to rise and fall while still maintaining head pressure on the system pump. At the same time, the expansion tank acts as a “burp” tank, allowing for air in the system to escape.

If any component, such as a fan heater, sits higher than the expansion tank, it will (essentially) take over that function. Air will travel to the highest point in the system and stay there. For example, if a defrost fan heater were installed above the expansion tank, air would trap in the fan heater and it would most likely not perform to specifications. Further, the air would bind the water flowing through the fan heater and plug the loop in which it is installed, leading to failure of other components in the system.

The expansion tank may be installed anywhere in the system provided it is the highest point and there is no restriction between it and the system pump. It is possible and in most cases desirable to install the expansion tank on one of the loops from the manifold. The loop with the expansion tank should be marked to warn against shutting it off completely. One side can be closed to stop circulation but still allow for expansion.

## Water Conditioning/Filtration

A water filtration system is available. It consists of a filter and parts necessary to mount and plumb it into your system. The filter contains an alkaline material that helps maintain the proper PH of the coolant.



## System Circulation Pump

For proper operation 90 degree fittings must not be mounted on or near either suction or discharge sides of the pump.

The system pump must be free from air for good circulation and proper water flow through the system. Air bubbles in the pump can cause cavitation that will impede the flow of water out of the pump. Many factors can lead to pump cavitation: improper installation, air trapped in components such as fan heaters, or improper seals in the water system.



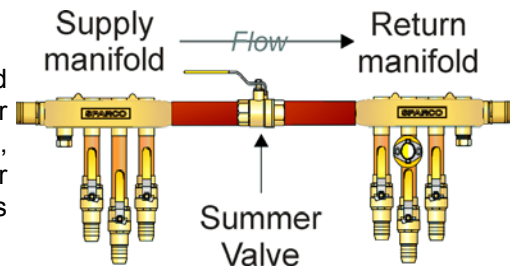
## Supply/Return Manifolds

A manifold is used to branch the main water line into the various zones. The supply manifold feeds the heated coolant to the various zones and the return manifold brings the zones together and returns the coolant to the pump. All valves in the system should be full-port valves that do not restrict the flow of water through the system. SMS offers a manifold kit that can meet all system requirements.



## Summer Valve

During the summer you may want to use a "Summer Valve" to avoid circulating hot water through your heating system while making your domestic hot water. Instead of having to shut all valves in a manifold, a bypass (Summer) valve will allow water to pass through your domestic water heater and not through the cabin heating loops. This allows you to make hot water without heating the boat.



The summer valve is a bypass valve installed between the supply and return manifolds. During regular heater operation, the valve is shut. When the summer valve is shut, water flows throughout the entire system. If the summer valve is opened, water follows the path of least resistance and flows directly through the supply and return manifolds, bypassing the cabin heating zones.

## Domestic Water Heaters

A domestic water heater should be on a loop that will always be on. If the domestic water heater is installed in a zone that is shut off for any reason, it will no longer have the ability to be heated by the boiler.

An exception to the rule that nothing may be installed between the expansion tank, pump and heater is the Everhot brand domestic water heater. This water heater provides minimal restriction to the flow of the coolant through it.

**A diesel boiler can provide VERY HOT water.**

Whenever using a diesel boiler to heat domestic water it is necessary to install an Anti-Scald valve (also known as a thermostatic mixing valve).



**Failure to do so may cause serious injury.**

## Fan Heaters

Fan heaters should be spaced throughout the boat based on the BTU demands of each area to be heated.

All living spaces should be heated to ensure an even, efficient heat for your boat. An unheated room will draw heat from the system unless it is always sealed from heated areas. This can cause drafts and result in uneven heating.

To figure the BTU demand of a space, measure the volume (Length x Width x Height) taking in to consideration the contour of the boat. Once volume is figured, multiply the cubic feet by 12 if the space is below deck, or by 15 if the space is above deck or has many windows. This figure should give you an idea of the BTU demand for this space.

Once the BTU demands are figured, fan heater models can be chosen. In most cases, room for fan heater installation guides the choice as much as the heat demand. A single unit may meet the demands for a space, or several units may be required. Further, when one unit may suffice, there may not be enough room for its installation and multiple smaller fan heaters may be needed. When choosing where to put the fan heaters, ensure that the heat will be ducted into the space evenly. If a space requires 24,000 BTU, try not to install every fan heater in one location, arrange them to evenly heat the boat.

As you select the fan heater locations be aware of where the hose must run, and in which zone you want the fan heater. Also, be sure that the location will meet the manufacturer's installation requirements. To avoid spreading dangerous or noxious fumes all fans should draw either from the cabin or outside air.

## Thermostat(s)

Mercury thermostats are not to be used, they are easily affected by motion and vibration. They will cause short cycling and intermittent running of the heater. If you decide to use a thermostat other than the one we have provided, contact SMS to find a compatible model.

These thermostats are low voltage, low current devices. They can signal a relay to power the fan heaters.

When deciding where to mount the thermostat, we advise temporarily wiring it on an extended cable, and testing it in various locations to find the best location for proper heating of your boat. This will help you find the spot that is least effected by things like drafts and direct sun which can fool the thermostat.



REAL 6000



REAL 843x



Aube TH135



REAL 6400

## Exhaust System

The exhaust must be piped overboard. It is a high-temperature dry exhaust that requires insulation. The system consists of rigid exhaust tube bent to necessary angles to install in your boat, an exhaust thru-hull, exhaust clamps, exhaust sealant and insulation blanket.

For the purposes of this manual, installation of the exhaust will begin by running the exhaust to the hull, installing the thru-hull and finally sealing the exhaust and wrapping with insulation blanket. In some cases, however, it is required that the thru-hull be installed first and the exhaust is installed back to the heater.



**Maintain at least 1" clearance from the outside of the exhaust insulation to wood or ANY other combustible material.**

This unit utilizes a 1½" exhaust system. Components must be made of type 316 stainless steel to ensure a long life.



**The exhaust run may not exceed 270 degrees in bends and 10 feet in length.**

Elbows may be at any place in the system and in any order, but no elbows tighter than 90° may be used.

To help avoid trapping condensation the exhaust system must exit above the boiler. A negative rise may cause exhaust gas backup into the boiler.

A dry place must be found for the installation of the exhaust thru-hull. Water intrusion into the exhaust system may lead to corrosive failure of the system or heat shock that can lead to cracks in the piping.

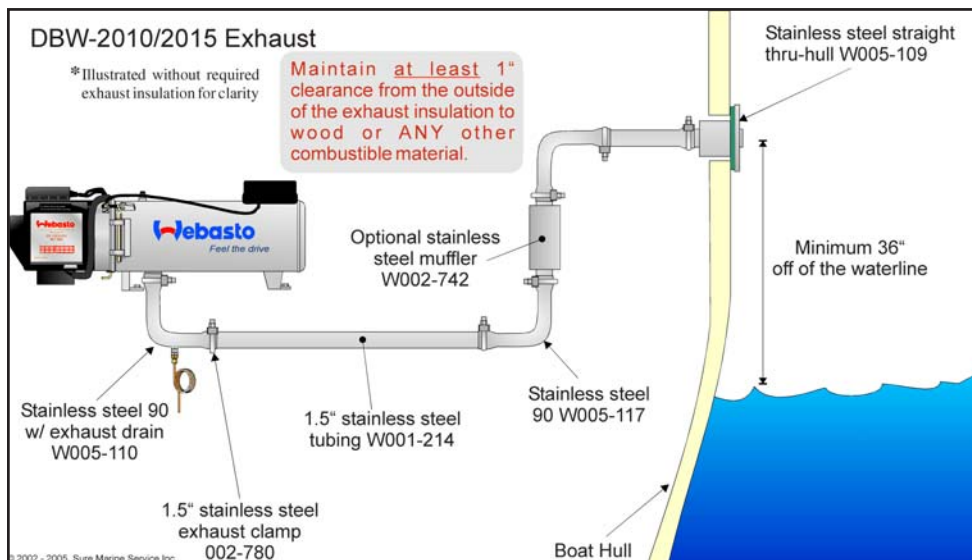
The most successful location for the thru-hull has been the aft transom. This location is usually protected from the rolling motion of the boat that would put the fitting in jeopardy from waves and rolling seas.

Due to space and component restrictions, the transom may not be available. If a side of the boat is used, the thru-hull should be placed as high and as far aft as possible. The higher and farther aft a thru-hull can be placed, the better protection it may get.

The thru-hull must be installed at least 36 inches above the waterline for proper operation.

If the thru-hull is located on the side of the boat, be aware of any fire hazards that may exist. Be aware of your exhaust when rafting to boats or docking. Avoid aiming the exhaust at flammable materials because you can cause them to heat up, melt, or ignite.

Once a spot is chosen for the thru-hull, the exhaust run should be mapped out. Exhaust lengths can be custom cut and an assortment of prefabricated 90- and 45-degree angled fittings are available. If the exhaust cannot be assembled from prefabricated pieces, custom parts can be made to fit.



An exhaust muffler may be used to quiet any combustion noises.

To avoid condensation from soaking the muffler packing, the muffler must be installed vertically.

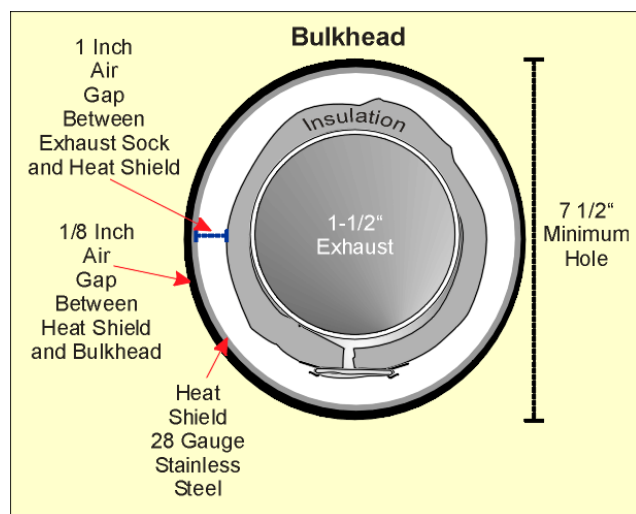


We have provided a "CAUTION Hot Exhaust" sticker. Attach it to the exterior near the thruhull to provide a warning.

When mapping the exhaust run, be sure that the exhaust will be properly isolated and insulated from combustible materials. The exhaust must not touch a bulkhead directly. Metal standoffs must be used to ensure stability of the exhaust system and to provide isolation from combustible materials. High temperature insulating blanket must be installed as needed.

## Bulkhead Insulation

In many installations, the exhaust must pass through bulkheads to reach the thru-hull. To prevent heat transfer from exhaust to bulkhead, insulation must be wrapped around the exhaust (see below) and any hole cut for the exhaust must include a suitable air gap.



When running the exhaust and insulation through a bulkhead, a guard (thimble) must be installed. This thimble will be a heat shield between the exhaust and the bulkhead.

When installing the thimble drill the hole 1/4" larger than the diameter of the thimble. You should have an 1/8" air gap between the thimble and bulkhead. The exhaust should not touch the thimble and should have a 1" air gap between the insulation and the interior walls of the thimble. The thimble should be constructed of at least 28 gauge stainless steel. The thimble should extend 3" from either side of the bulkhead, but must not exceed 8". Heat resistant gaskets should be installed on the flanges of the thimble. No part of the thimble should come into contact with the exhaust.

Watertight bulkhead fittings are available by special order.

## Trial Fit

Once all the bulkheads have been drilled, piece the exhaust together to the point where the thru-hull will exit the hull.

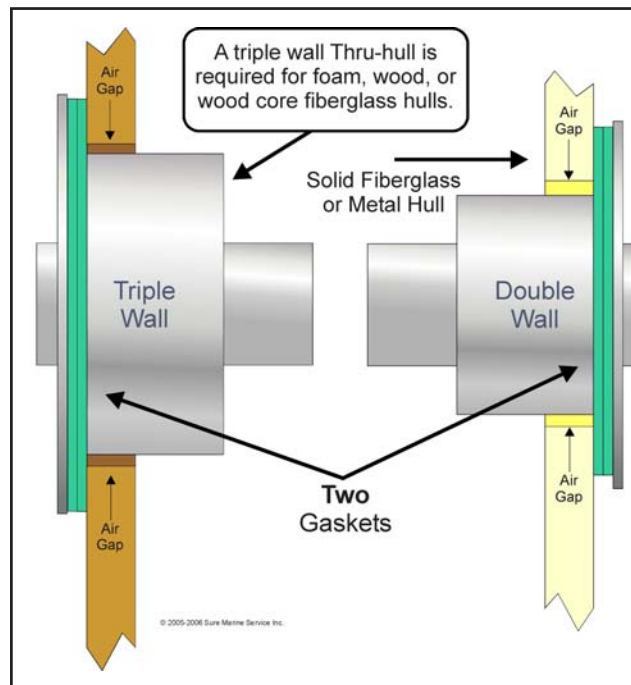
### Do not seal the exhaust yet!

Adjustments may be needed after the installation of the thru-hull and your sealed exhaust system would have to be taken apart.

## Thru-Hull Fitting

At this point, the hole for the thru-hull should be measured and cut. Measure and mark carefully because you will be drilling a hole through the hull of your boat.

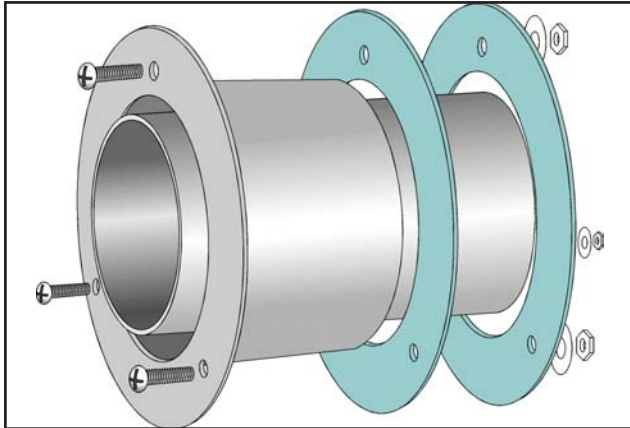
You will be using a double or triple wall thru-hull depending on the heat isolation requirements of your hull.



If a triple wall thru-hull is used, make sure there is clearance for the nuts on the back of the thimble.

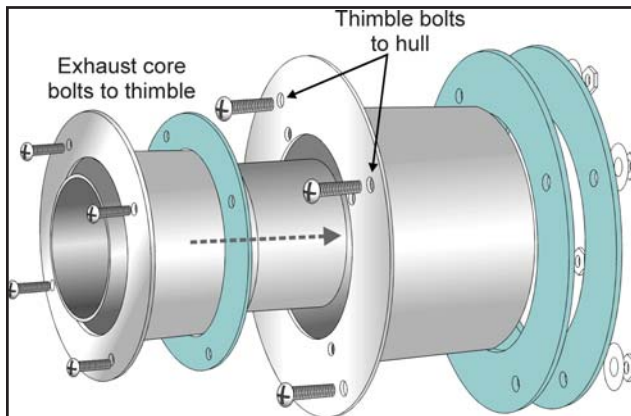
Before drilling the mounting holes, ensure that when the thru-hull is centered a 1/8" air gap is maintained around the body of the thru-hull.

Once you are sure of a proper fit and air gap Mark the holes for the fasteners and drill guide holes for the mounting screws.



## Triple Wall Thru-hull

The thimble must be bolted, not screwed to the hull. Machine screws will be inserted through the thimble, washers, and hull. Washers and nuts on the inside of the hull will secure the thimble. This is usually a two person job. Again, ensure a 1/8" air gap exists evenly around the thimble.



## Gasket Sealer

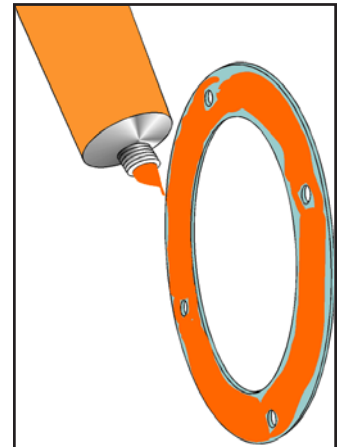
The thru-hull must be installed with the provided insulation gaskets to prevent heat transfer to the boat's hull.

Before mounting the gaskets on the thru-hull, liberally apply a coat of high temperature gasket sealer to both sides of each gasket.

The coated gaskets are placed over the thru-hull and the assembly is inserted into the hole you drilled.

As you tighten the mounting bolts make sure you maintain the 1/8" gap around the thru-hull.

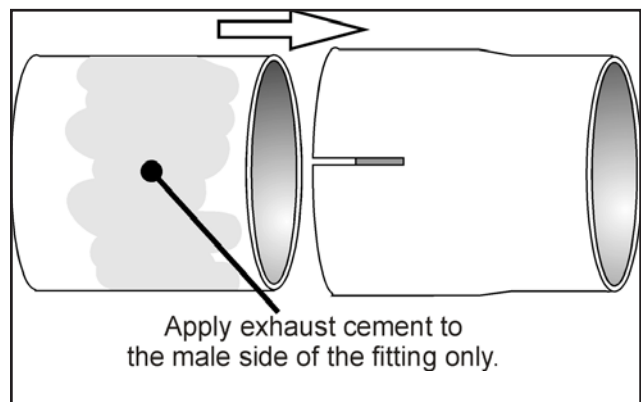
Snug up all screws or bolts evenly and check to see you have mounted the thru-hull firmly to the hull.



## Seal the Exhaust

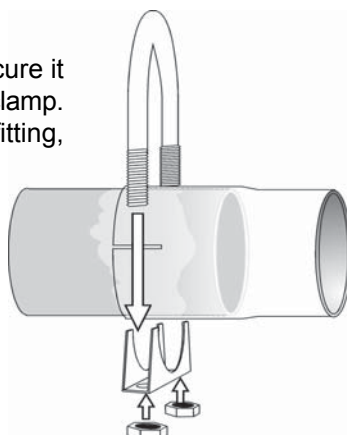
Once the thru-hull is installed, the exhaust system can be sealed and permanently installed.

All fittings in the exhaust system must be properly sealed and clamped to prevent any leakage of exhaust gasses into the boat's living spaces.



When applying the exhaust sealant, it should be applied to the male side of the fitting only. If the sealant is applied to the female side, it may impede the flow of exhaust gasses and cause problems with the heater.

Once the fitting is sealed, secure it with a stainless steel exhaust clamp. Slide the clamp around the fitting, over the slots in the bell, and tighten evenly. A loose clamp may lead to exhaust leaks even though the exhaust sealant has been applied.



Triple Wall Thru-hull

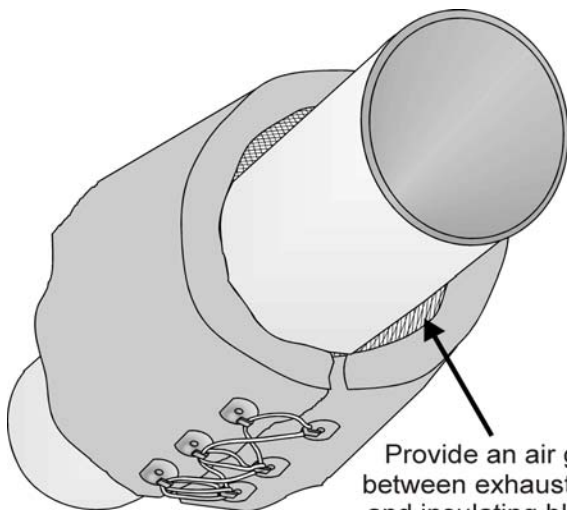


**Maintain at least 1" clearance from the outside of the exhaust insulation to wood or ANY other combustible material.**

Now that the exhaust system has been permanently installed, the exhaust insulation blanket can be fitted. Sure Marine Service provides a monel-screened, siliconized, high-temperature exhaust blanket that should be wrapped around the exhaust to help protect any objects that may come in contact. After wrapping around the tube, the blanket is secured by a copper or stainless steel wire.



High Temperature Exhaust Blanket



Provide an air gap between exhaust pipe and insulating blanket

The blanket is designed for a loose fit so that air between the blanket and the exhaust will provide insulation. Do not over tighten the blanket for a snug fit. This can lead to damage to the blanket, possibly to the exhaust tube, and decrease the value of the insulation.



## Fuel System

The boiler uses a double (supply and return) system .

A secure, airtight fuel supply system is required to ensure proper operation. The line should be Type A USCG-approved rubber fuel hose or heavy wall copper tube. All fuel lines and components should be a true 1/4" inside diameter. Restrictive components may affect the operation of the unit. All fittings in the system should be accessible for maintenance and inspection and should be sealed with fuel-rated pipe thread sealant to prevent leaks.



**Teflon tape should never be used to seal a fuel system, it will void your warranty. Most tape is not rated for fuel and small pieces may enter the system and foul the burner.**

A fuel supply or return line should never tee into an existing fuel line, such as an engine fuel line. A supply line tee may starve the fuel pump if it must compete with another pump. Backflow from a return line tee may damage the fuel pump and ultimately the heater.

Per ABYC standards, a shut-off valve must be installed at the tank on the suction line.

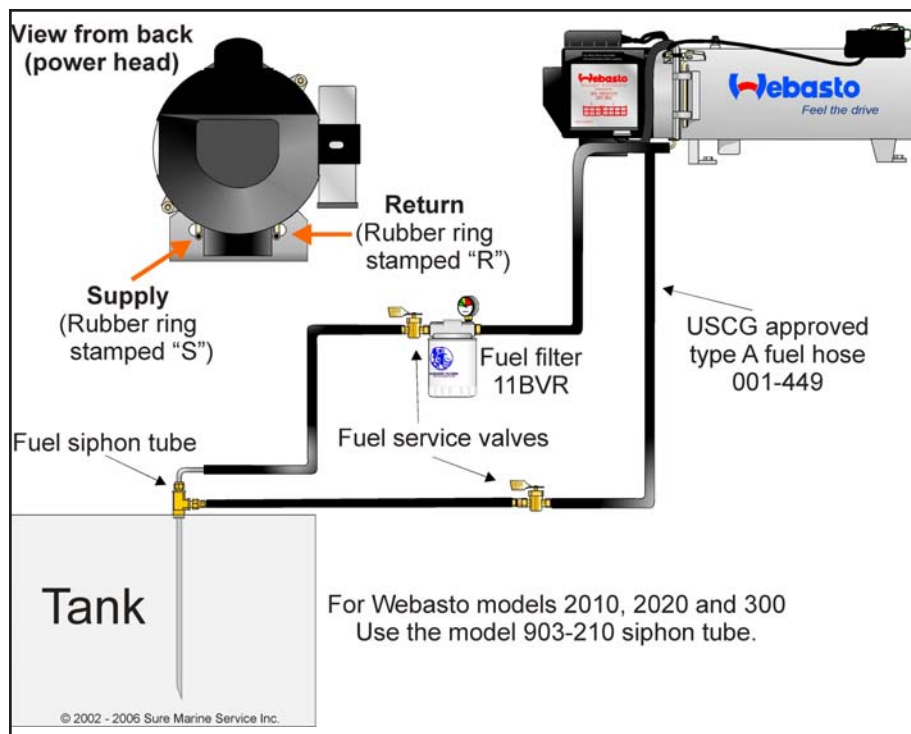


Isolation valves should also be installed on both the supply and return sides for future maintenance of the heater. These valves should be free-flow valves that do not restrict the fuel flow through the lines.

Fuel lines must run to and from the same tank. Supply from one tank and return to another can lead to a fuel transfer that may lead to fuel spills, burner flameouts and damage to the boiler.

The return line must terminate 3 to 4 inches above the supply side. Failure to do so may lead to air in the fuel system and possible flameouts.

The length of the fuel lines limit the maximum lift the pump can handle. Since there is a supply and a return line, the length of the run is double the distance between the boiler and the bottom of the fuel tank. Webasto lists a maximum run of 32 feet (16' each way) with 6.6 feet of lift.



The Heater can be supplied from an auxiliary diesel fuel tank. This way a gas powered vessel can take advantage of diesel heating.



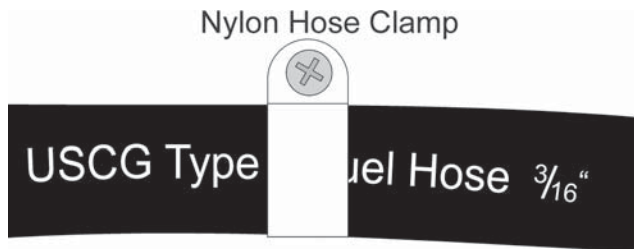
SMS stocks 2, 3, and 5 gallon aluminum diesel fuel tanks.

## Installing The Fuel Lines

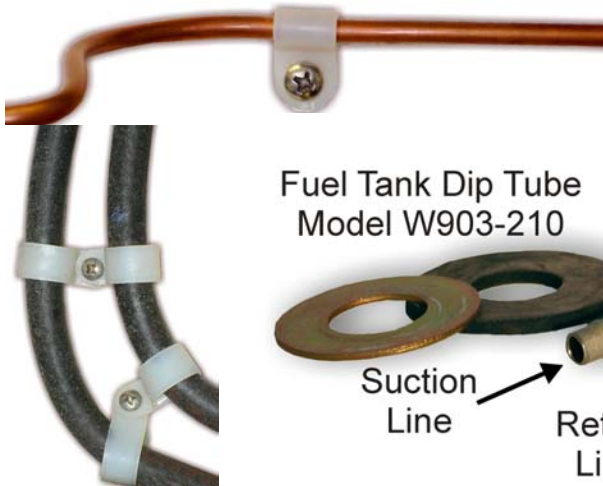
As you install the fuel line mount all components permanently assuring they are liquid and vapor tight. All hose connections must use corrosion resistant Stainless Steel clamps over the hose barb.



Per ABYC standards, all fuel lines should be supported by clamps attached to the bulkhead with corrosion-resistant fasteners and of a design to eliminate cutting or abrading action where in contact with the fuel line.



To further support the fuel line, a fastener should be installed every 12 inches. SMS offers nylon clamps designed specifically for fuel hose or metal tube.



Each component of the system must be independently mounted. The fuel system cannot support the weight of components such as filters or valves.



## Fuel Filter

Due to unpredictable quality of fuel and the possibility of foreign matter entering the burner, a fuel filter must be installed in the fuel supply.

The injector nozzle system, like any other diesel injector system, can be fouled by foreign matter potentially causing damage to the burner or the boiler.



SMS recommends the model W11BVR filter.

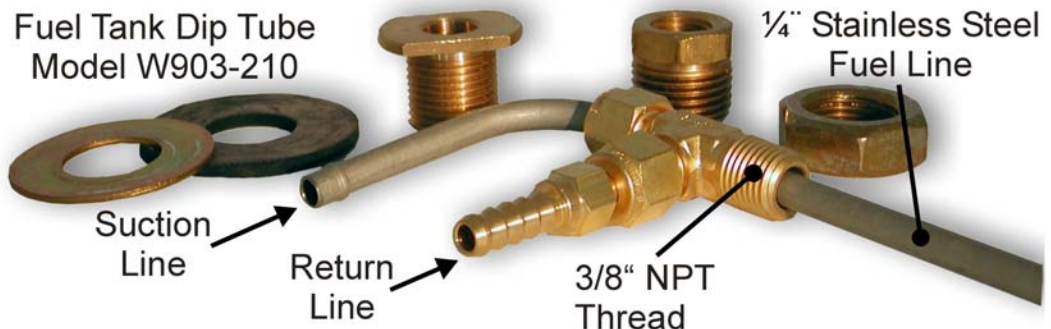
The Garber meets all ABYC standards and can be fitted with a vacuum gauge to indicate when to change the cartridge.



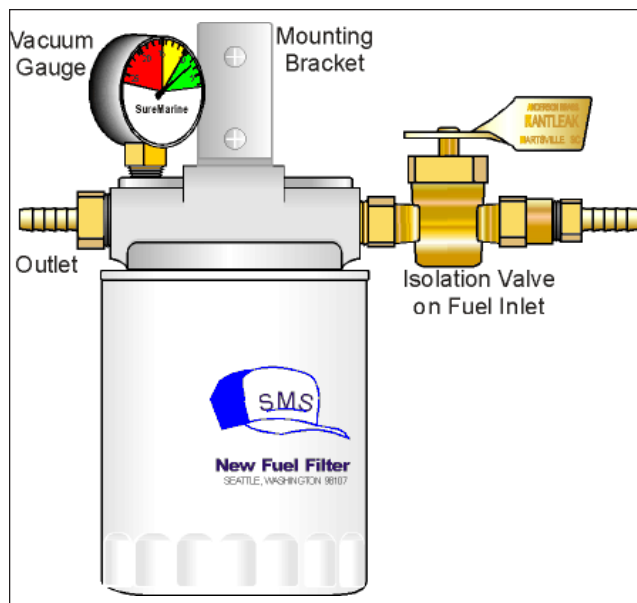
The fuel filter should be installed with an isolation valve to assist in changing the cartridge.



Fuel Tank Dip Tube  
Model W903-210

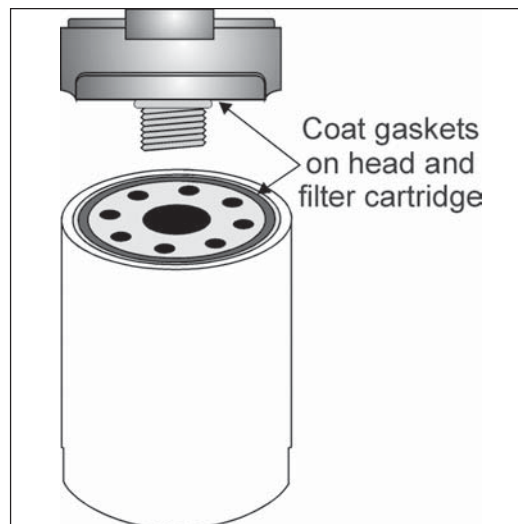


Like any other component in a fuel system, the fuel filter must be mounted so it is accessible for changing and maintenance and it must be independently installed on a bulkhead.



When deciding on where to mount the filter, make sure the space allotted has enough room for all parts of the filter and hose fittings as well as enough room below the filter so the cartridge can be unscrewed and removed for replacement. About 2 inches should be sufficient.

Before tightening the filter cartridge on to the head, spread a thin layer of petroleum jelly or fuel on to the gasket on the cartridge and the O-ring on the head. A proper seal may not be made with dry gaskets and air may seep into the system potentially causing flameouts and damage to the heater and fuel pump.



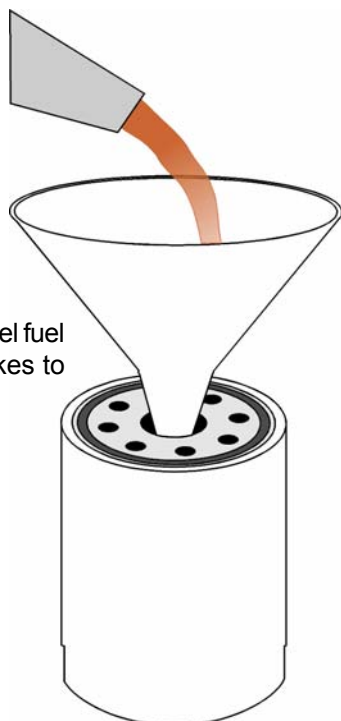
Once all the steps for the installation of the filter are completed, screw the filter cartridge on to the filter head.

Tighten the cartridge  $\frac{3}{4}$  turn past when the gasket on the cartridge touches the head.

Overtightening the filter cartridge can lead to damage to the gasket on the cartridge and can lead to problems with removal later.

Under tightening the cartridge can lead to air seeping into the system and potential flameouts and damage to the pump.

Prefilling the filter with diesel fuel will shorten the time it takes to bleed the fuel system.



## Electrical System

All wiring and electrical connections should comply with the ABYC, Coast Guard, National Electrical Code (NEC) and with local codes and practices.



Due to the possibility of stray current galvanic corrosion, a #10 gauge wire should be installed on the heater and water pump leading to the boat bonding system to prevent such corrosion, as per ABYC standards.

We recommend #14 gauge for thermostat and fan heater wiring.

All wiring should be safely secured with wire ties and tie downs. Ensure all parts of the electrical system are installed to approved standards and workmanship.

### SureWire™

SMS has provided a SureWire board which standardizes the wiring connections simplifying installation and troubleshooting. Additionally the SureWire provides a system power LED, pump in operation LED, priming button, and an LCD hour meter. The SureWire board must be installed in a clean, dry and accessible location.



**The wire harness from the SureWire board to the heater must not be modified.** SMS will custom make a longer wire harness if required.

### Wire and Power

The rated voltage must be available at the heater connections when it is under peak load. Though wire size may not seem like the most important choice when installing a heater, poor wiring will cause problems. The heater draws a lot of current (control system, ignitor, and pump) during start up. During peak load marginal wire or poor connections may cause the voltage reaching the heater to be too low to ignite the fuel. Continued running with low voltage will cause the fuel pump and fan to slow resulting in poor combustion and/or over heating.

When measuring wiring include the complete supply and return run. The negative and positive are separate parts of a run. In general, #10 wire is advised for distances up to 20' total. The voltage drop at peak load should be less than 1/3 of a volt. If you require a longer run of wire, a larger gauge is recommended. Check with your electrician or contact SMS for advice.



**Your heater must never be shut down by the main power breaker.**

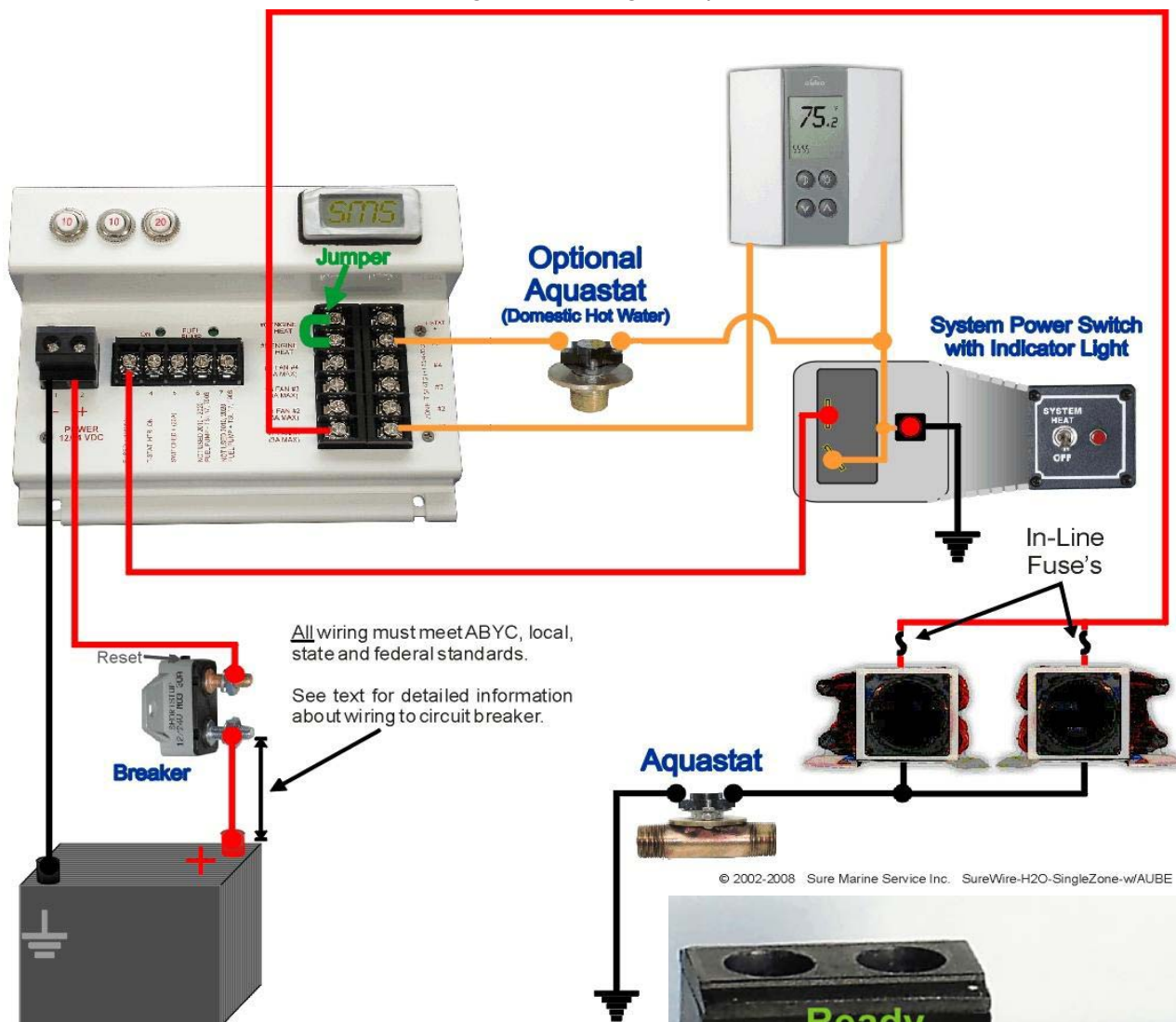
The unit must go through its 90 second cool down before shutting down. If the heater is shut down by the main power breaker, and the unit cannot go through its cool down, components that rely on airflow for cooling will be ruined from an over heat. The least possible damage will be the opening of the overheat circuit. This will result in a fault in the heater and the unit may not start the next time heat is called for.

When installing the main wiring from the battery to the SureWire board, a 30 amp breaker (approved type provided in kit) will be installed on the positive side. According to ABYC standards, the breaker should be installed within 7" of the connection to power unless the wire is protected by a sheath, then it should be within 72" for batteries or 40" for other sources of power, such as a battery switch.

Do not use plastic automotive-style fuses for the installation of the unit. These fuses will only fail in the long run and may cause more money than they save.

All wiring should be safely secured with wire ties and tie downs. The heaters main power wires should not be installed next to large amperage AC or DC wiring such as radar, sonar or refrigerator. Surges in these wires can affect the operation of the heaters control unit.

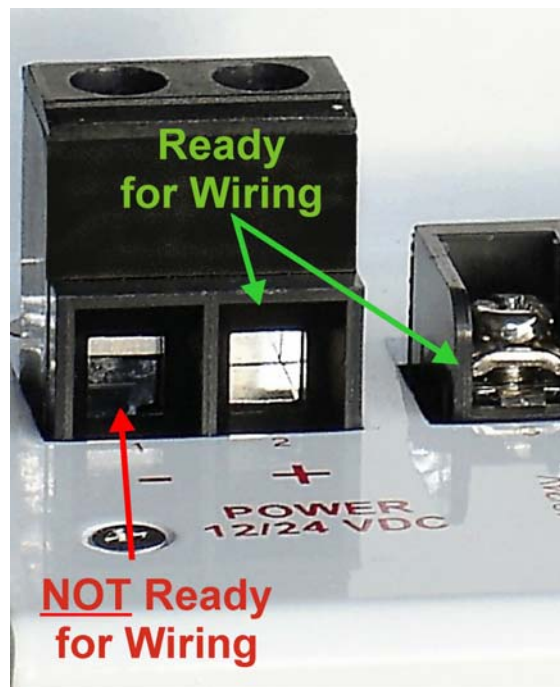
## Single Zone Wiring Example



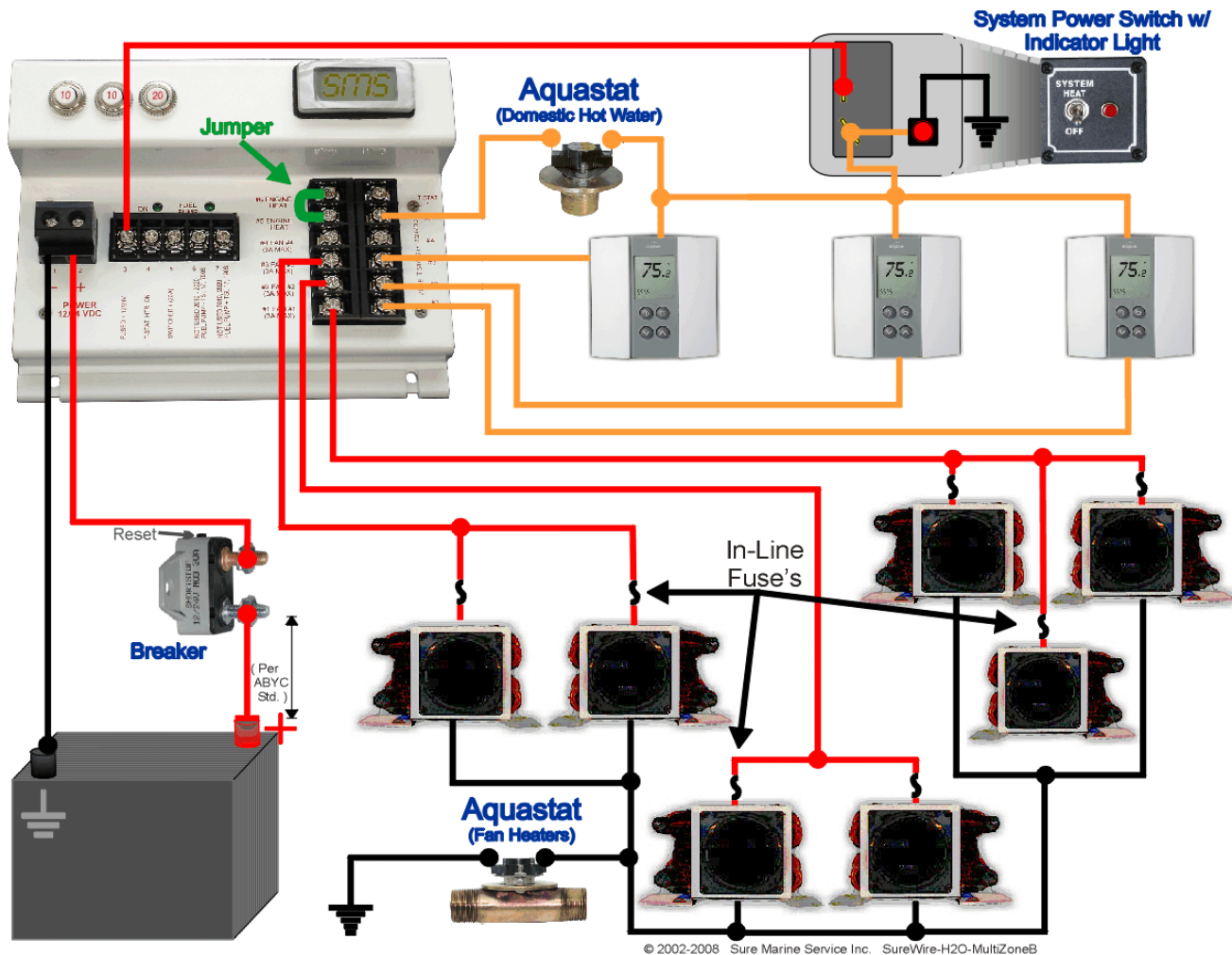
There are 3 barrier strips on the SureWire board for connecting your wiring. To avoid accidental damage from shorts, finish all control wiring before you make the battery connection.

The battery supply uses the heavy duty barrier strip (Labeled 1-2). This barrier strip will accept wire from 6 to 18 gauge. Check with your electrician or contact SMT for advice on wire size. Open its wire clamp by turning the screws counterclockwise, insert the wire, then secure by turning the screw clockwise. Connections to the shorter, wider barrier strips are made by unscrewing to raise the metal clamp, placing your wire under the clamp and screwing the clamp down to secure the wire.

Maximum torque for terminal 1 and 2 is 18 inch pounds.



## Multi Zone Wiring Example

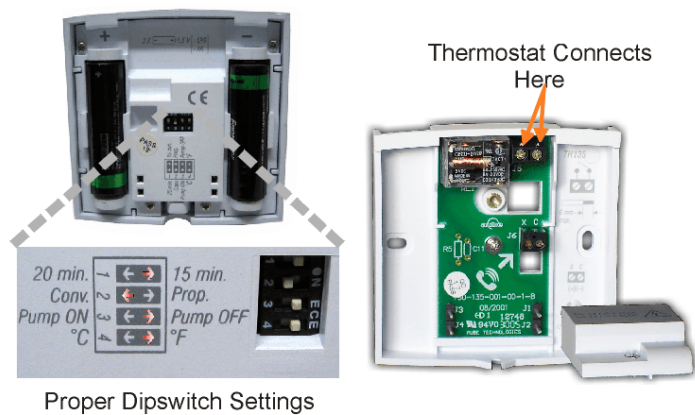


### Thermostat Connections

Commonly thermostats made for the U.S. will have a connection marked either "R" or "RH" for power and another marked "W" for the heating signal. In a heating only system you will not be using any connections marked "G", "Y", or "RC".

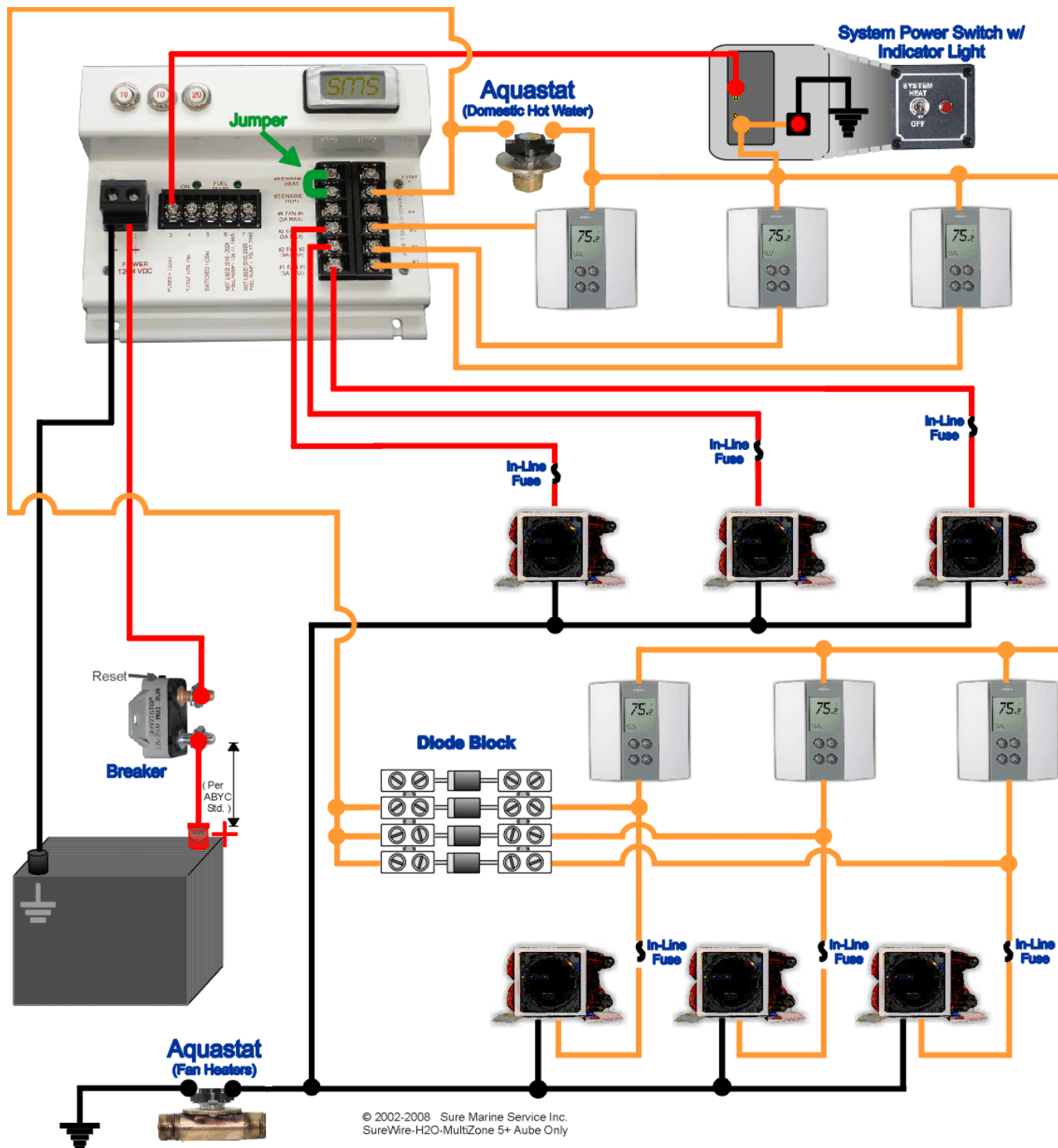
European thermostats such as Webasto or Aube may have different markings. Please consult their instructions.

### Aube TH135 Thermostat Detail



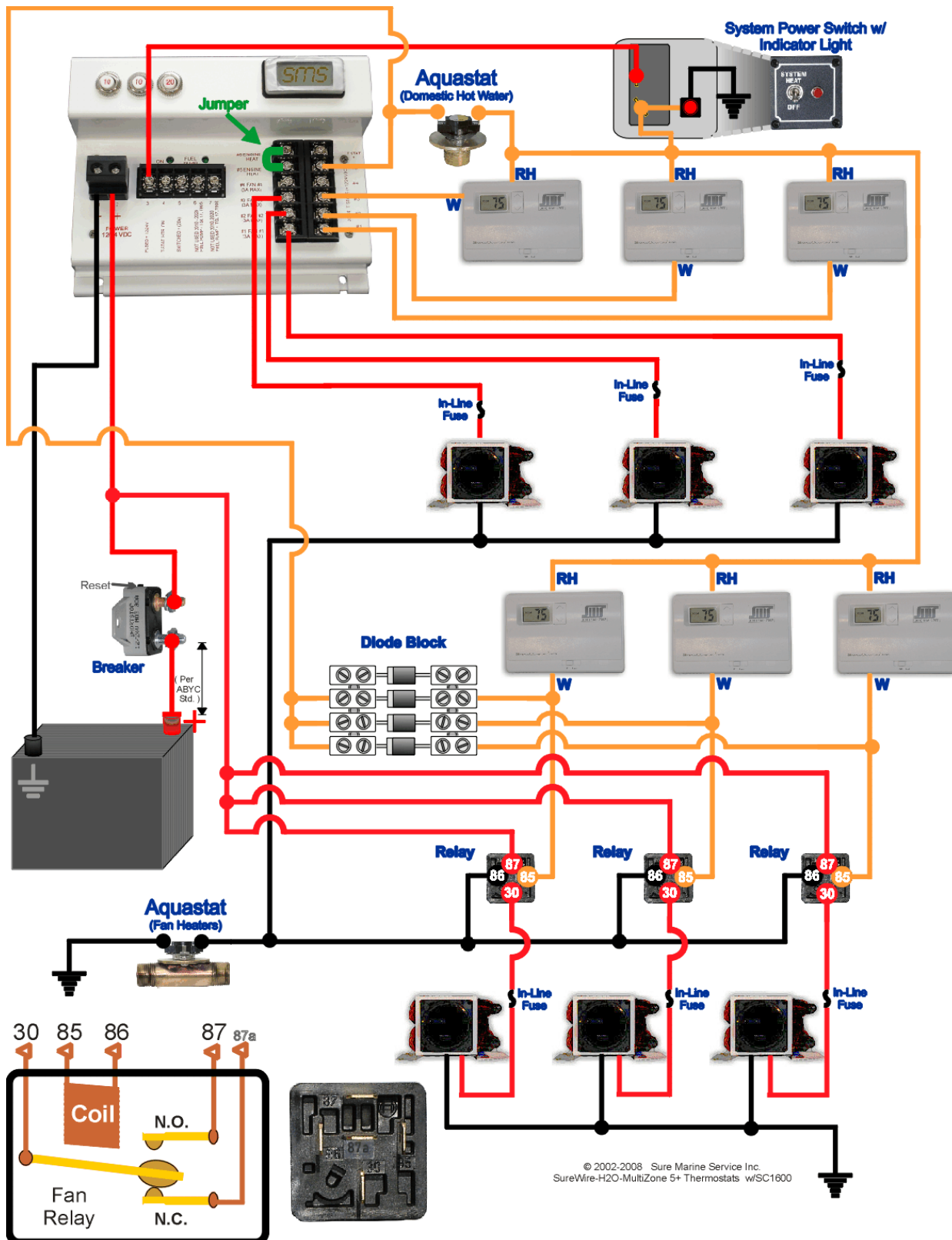
## Multi Zone Wiring Example with more than 4 zones using Aube thermostats

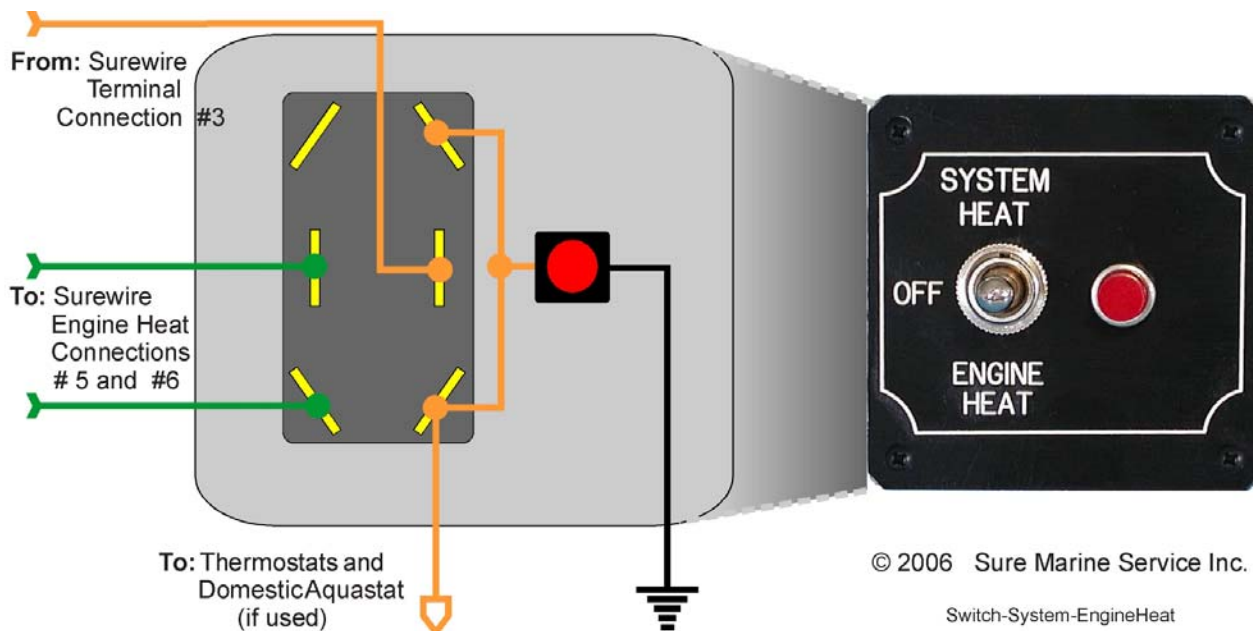
The Surewire™ wiring board can be used to wire up to four zones of thermostats. If more than four zones are desired a diode block must be used. The Aube thermostat has a built in relay and can accept up to 2 amps of fans. If a thermostat other than an Aube thermostat is used relays must also be added to the system. See wiring diagram on the following page.



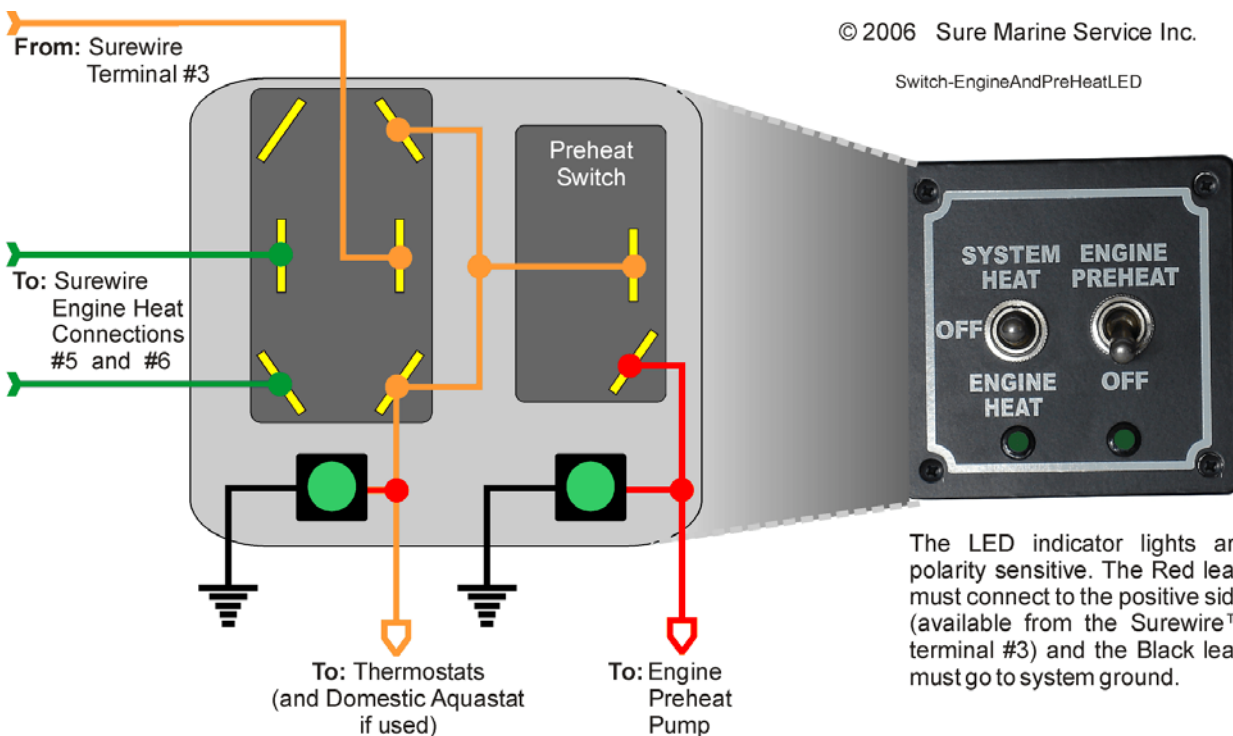
© 2002-2008 Sure Marine Service Inc.  
SureWire-H2O-MultiZone 5+ Aube Only

*Multi Zone Wiring Example with more than 4 zones using SC1600 thermostats and relays.*





The factory installed jumper across terminals #5 and #6 on the Surewire™ **MUST** be removed to allow "Engine Heat" switch operation.



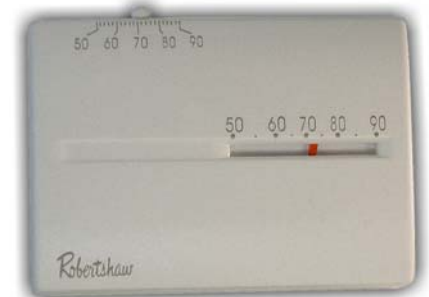
## Thermostat(s)

Room thermostats must have snap action contact points to prevent point bounce during rolling conditions. The milli-volt type of thermostat MUST NOT be used.

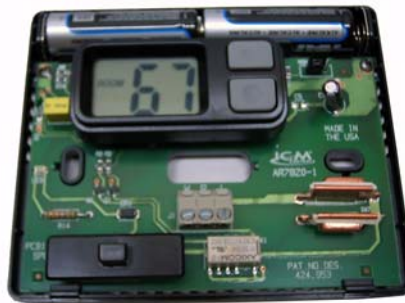
Mercury thermostats are not to be used, they are easily affected by motion and vibration. They will cause short cycling and intermittent running of the heater. If you decide to use a thermostat other than the one we have provided, contact SMS to find a compatible model.

These thermostats are low power signaling devices. The SureWire™ board controls the boiler and can manage 3 amps per heating zone. Use relays to control higher current or higher voltage devices.

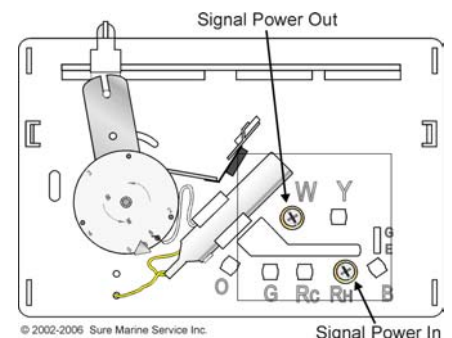
When deciding where to mount the thermostat, we advise temporarily wiring it on an extended cable, and testing it in various locations to find the best location for proper heating of your boat. This will help you find the spot that is least effected by things like drafts and direct sun which can fool the thermostat.



Aube TH135



SC1600B



W001-467

## Aquastat

To prevent cold air discharge from the fan heaters during initial startup, an inline temperature sensing aquastat is available to install in the return line. This aquastat keeps the fans off until a water temperature of 120° F has been reached.

## Circuit Breakers and Fuses

The main breaker is to protect the main power line, not the rest of the system.

Each fan should be protected by an appropriately sized fuse or breaker.

## Fan Relay

A fan relay allows a low power signaling device, in this case a thermostat, to control higher current or higher voltage devices. Each of these relays can operate one or more fans. This general purpose has contacts rated to 30 Amps at 12 VDC. A socket with spade connectors is available for this relay.

## Master Switch

A “Master Switch with Indicator Light” allows you to quickly and clearly determine if your heating system is enabled.

This satisfies the ABYC requirement for a positive shut off with a visible system status indicator.

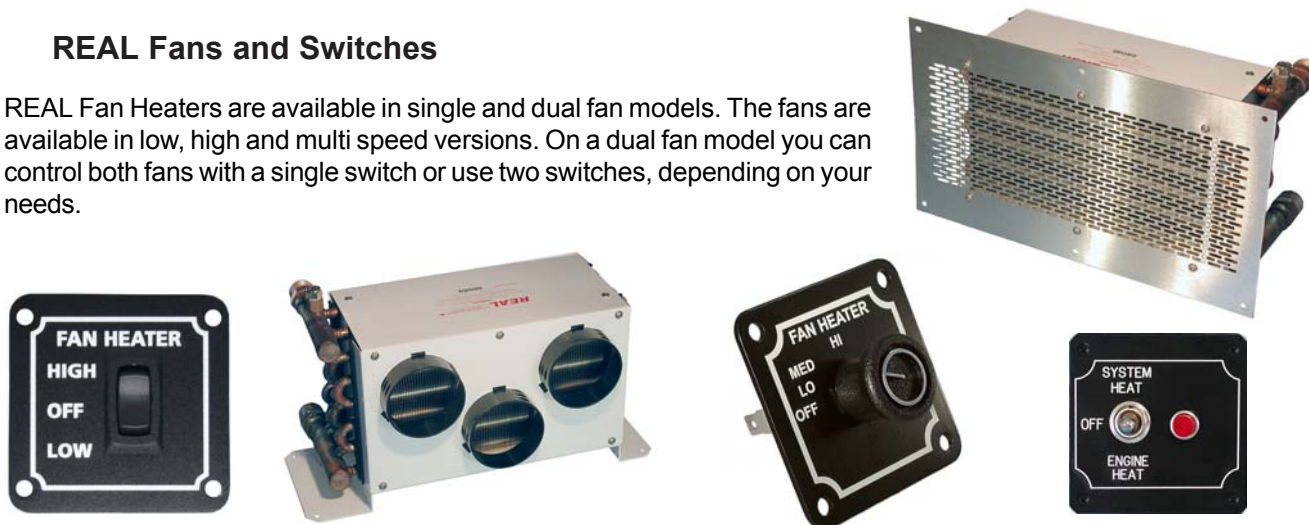


Place the provided “IMPORTANT” warning sticker near the heating system master switch.

Optional switch plates labeled SYSTEM HEAT/ OFF/ ENGINE HEAT and SYSTEM HEAT/ OFF / ENGINE HEAT / ENGINE PREHEAT including an indicator light are available. Contact SMS for wiring and plumbing diagrams.

## REAL Fans and Switches

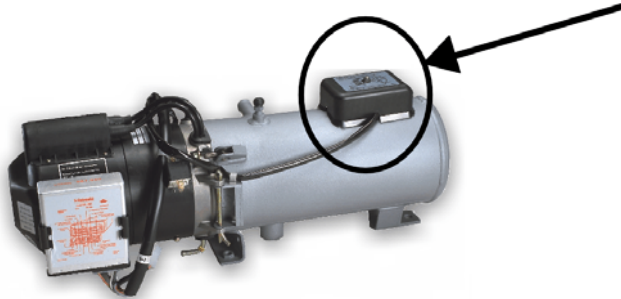
REAL Fan Heaters are available in single and dual fan models. The fans are available in low, high and multi speed versions. On a dual fan model you can control both fans with a single switch or use two switches, depending on your needs.



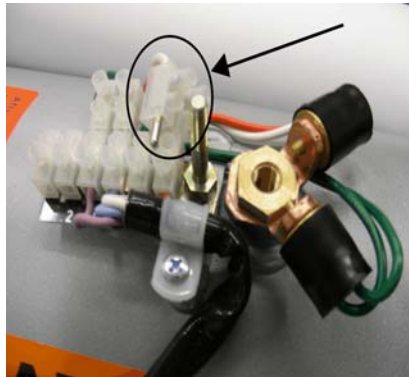
## Control Thermostat



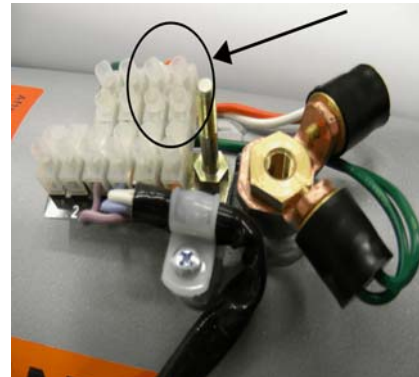
**This unit ships with the control thermostat disconnected. The unit will not fire until control thermostat is connected.**



The control thermostat is located below the black plastic cover on top of the heat exchanger as pictured above.



Control thermostat unplugged.



Control thermostat plugged.



**Do not connect control thermostat until all air has been purged from the system.**

## Control Box "B" Plug



**For your safety, disconnect all power to powerhead before opening heater.**

The heater has no way of knowing that it is open. The unit can fire when the powerhead is open. Disconnect the power to the powerhead prior to opening heater. One good way to do this is to disconnect the "B" plug from the control box. See picture below.



"B" plug connected to control box.



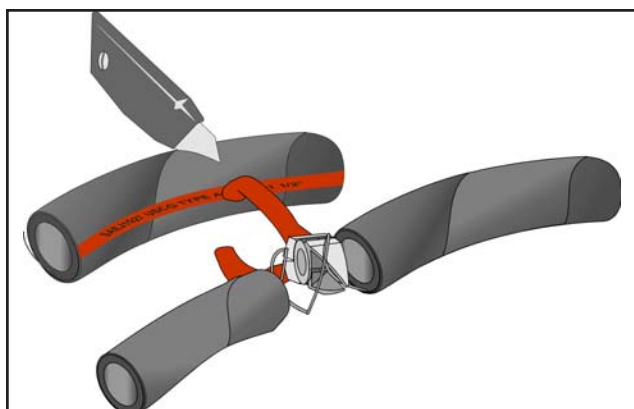
"B" plug disconnected from control box. Powerhead cannot fire and is now safe to open.

## Water System

### Hose

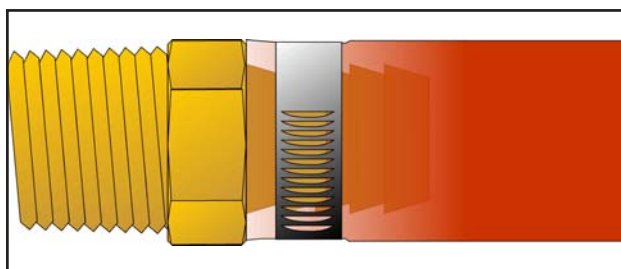
Rubber hose is the most common method used for the installation of hydronic heaters. It is flexible, easy to install and the proper hose can last the life of the heater. Proper hose choice in this matter is key. Standard water hose and common automobile radiator hose do not work for the long runs required. If the hose is not specifically designed for heater installation, it will tend to kink or fail causing damage to your heating system and possibly to your boat.

All hose should be reinforced rubber hose. This type of hose will not soften with the operating temperatures of the heater. The case of the reinforced hose can be cut with a knife but the wire reinforcement will require wire cutters.



All hose connections such as hose bars or unions should be made of brass, copper or stainless steel. All hose clamps used to join the hose should be stainless steel. Further, SMS advises the use of double-banded clamps. Double-banded clamps protect the hose from being cut by the holes in the clamp itself.

The hose should fit snugly and completely over the hose barb. The clamp should be tightened over the barb.



If you have to cut a hole to route a hose, the hole should be at least  $\frac{1}{4}$ " of an inch larger in diameter than the hose. This will allow a  $\frac{1}{8}$ " gap around the hose which will help avoid chaffing. This gap should exist even if a shield is installed for the hose to pass through.



The system hose should be supported by corrosion-resistant stainless steel or plastic clamp to a bulkhead or another stable surface. Nylon hose mounts are the most common and cost effective way to accomplish this.

If the system installation calls for a tight turn in the hose, a copper 90-degree elbow should be used instead of attempting to force the hose to make the tight turn. By forcing the hose, a kink may form and pinch off the system.



If the hose makes a high point and may trap air, an in-line bleeder or 90-degree bleeder should be installed.

A fill/drain valve with common garden hose thread should be installed at the lowest point in the system. This valve is to assist in the flushing, draining and filling the system for initial start up and periodic maintenance. Sure Marine Service offers this valve and any hardware necessary to install it into the system.

## PEX

PEX (Cross-linked Polyethylene) has been used in potable water and for hydronic heating systems in Europe for over 30 years. PEX has replaced polybutylene (PB) as the most widely-used flexible plumbing pipe. We use PEX that is specifically made for radiant heating.

It is unaffected by electrolysis and compared to copper, installation should need fewer fittings without a need for torches, solder, flux, and so on.



To make a connection you slip a metal band over the tube, insert a fitting into the tube, then crimp in place using a crimping tool. The connection is ready for use. You can inspect the connection using a simple "Go/No-Go" gauge.

## Copper Tube

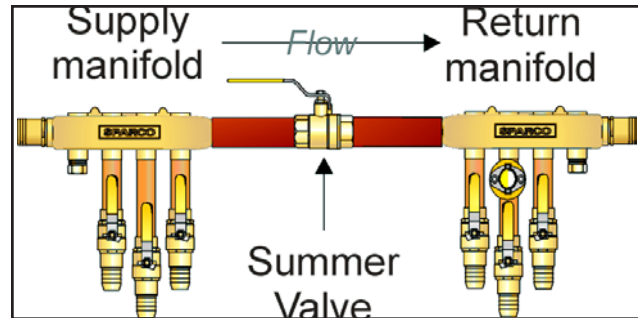
Copper tube is a standard for household heating but space restrictions limit its effectiveness in the marine environment.

Copper tube is used in installations that have long, straight runs. Because of the nature of boats in general, there are few long, straight runs of anything. Bends require soldering which, because of proximity to combustibles, cannot be done safely inside the boat. This requires the construction of complex pieces of tube outside the boat prior to installation. This is not the best method for a tight installation and many installers opt for hose or PEX tubing.

Copper tube can, however, be used in cases in which a small amount of heat should come from the system without the use of fan heaters or tube fin. Exposed copper tube is rated to 250 BTU per foot. An exposed length in a locker or lazarette can provide heat to dry clothes or gear. These exposed pieces should be protected and away from high traffic areas. Water flowing through these tubes is up to 180 degrees Fahrenheit and can burn exposed skin.

## Manifold

To increase the flow of the system and to ensure an even heat for your boat, a manifold distribution system is used. This takes the hot water supplied by the boiler and feeds it to several parallel heating loops.

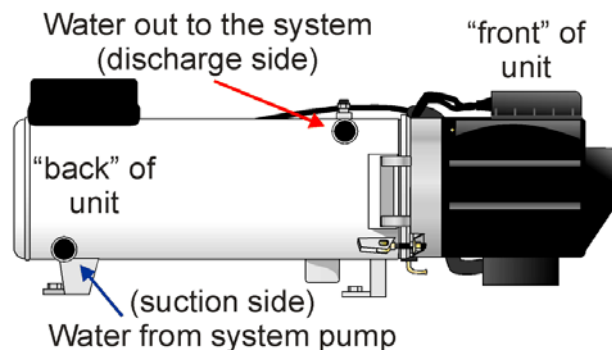


A summer valve lets you bypass the heating loops. This lets you provide heat to a domestic water heater without heating the boat and without shutting all loops. The summer valve is installed between the supply and return manifolds and is shut during normal operation forcing coolant through the heating loops. When opened coolant flows unrestricted through the main line only.

When installing a manifold, the heating loops should be installed facing down. This is to reduce heat convection into a loop that may not call for heat.

## Boiler

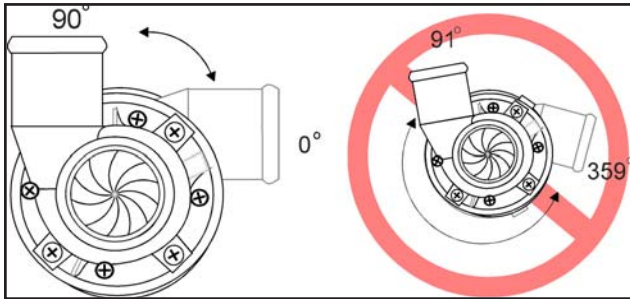
The boiler has two ports for plumbing connections, Supply (discharge) and Return (from suction pump). The lower fitting away from the burner head is the Return.



If the water outlet exits the heater and turns down a tee should be installed with a bleeder on top of the connection.

## Pump

Water flow through the system is very important and trapped air is the most common impediment to the proper flow. The system pump can be a source of air in a system if it is not properly installed. It is a centrifugal pump and it must have head pressure on the suction side to operate properly and it must be installed in a particular manner to move water without generating air bubbles. You are using water to distribute heat, bubbles won't work.

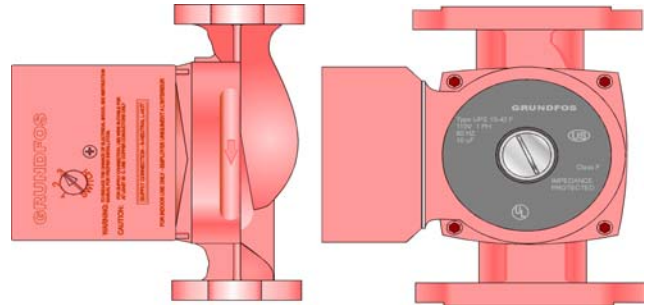


Most Centrifugal pumps are designed with center suction and side discharge. The side discharge must face between 0 and 90 degrees. Installation outside of this parameter will result in cavitation of the pump and put bubbles in the system.

The pump must be installed horizontally. If the pump is installed at an angle or vertically, it can trap air in the head that will adversely affect water flow



Not all centrifugal pumps are designed in this manner. The Grundfos (and similar pumps) have the suction and the discharge on opposite sides. These pumps must be installed either horizontally or vertically. Angled installation of any kind will result in air binding.

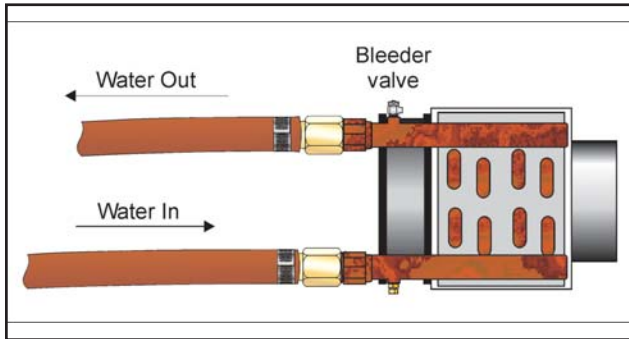


Consult the manufacturer's installation instructions or contact Sure Marine Service for any further pump considerations.

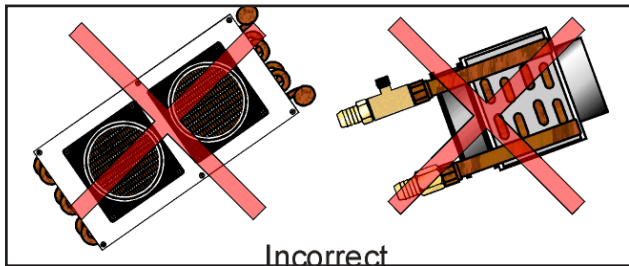


## Fan Heaters

Before installing the fan heater, make sure that the space selected will meet the manufacturer's specifications. Every fan heater should have a bleeder on the outlet side. If you can not locate an air bleeder, contact the manufacturer or SMS for options.



Water enters at the bottom connection and flows out from the top connection (unless otherwise specified by the manufacturer).



**No matter the make or model, a fan heater must be installed level. Some brands allow for several installation configurations, but all are on a flat surface. An angled installation will trap air such that it will not reach the bleeder.**

Once the system is installed and filled, the fan heater must be bled. This can be accomplished by opening the bleed screw on the discharge side of the fan heater and leaving it open until a steady stream of water flows out. If the water is spitting out of the fitting, air is still present. See the "bleeding the system" section.

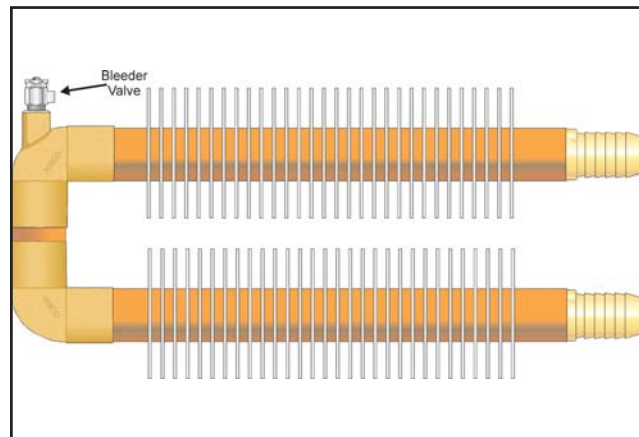


## Tube fin

Tube fin is a copper tube run through aluminum sheets designed to convect heat into the boat. This is the same system as a household baseboard heating system. Tube fin is advantageous because it does not draw any power from the batteries, however it requires considerably more room than a fan heater.

Most tube fin does not exceed 750 BTU per foot (3 x 3 inch) and commonly type is only 500 BTU per foot (2 x 2 inch).

While tube fin is convenient for small lockers and heads, several feet are required to heat larger spaces. Because of the angles of a boat, a long single run of tube fin is rarely feasible. Of course you can combine them with fan heaters in tight areas.



Two (but no more) lengths can be stacked on top of each other. If stacked tube fin is installed, a bleeder should be installed on the top tube.

Tube fin should be installed low in the cabin to allow for the heat to rise. A guard for tube fin should be fabricated to fit your set up. Standard household baseboard guards are available for single-pass tube fin, but for stacked systems it may be easiest to have a guard fabricated to fit.



## Domestic Water Heaters

The traditional domestic water heater is a tank of potable water heated by a loop from the heating system or by an electrical coil. Consult your water heater installation manual for mounting, wiring and plumbing specifications.

Most domestic water heaters have a 5/8" coil passing through the tank. A 3/4" bypass valve can be installed to allow water to pass through the tank while maintaining full heating system flow.

The bypass valve for the domestic water tank must be capable of throttling. SMS recommends using a gate valve for this purpose.

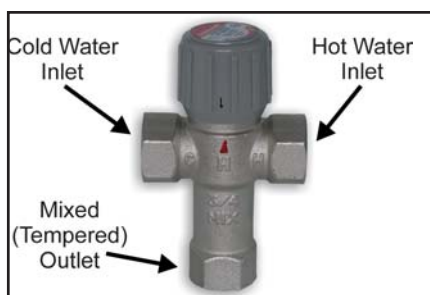


## Everhot Water Heater

The Everhot is an on-demand domestic water heater. Instead of a tank that stores water, the Everhot heats water by passing it through a coil of finned copper tubing surrounded by 180° F coolant from the boiler. Domestic water is heated up to 170 degrees F in one pass. The temperature rise is enough to require a mixing valve to avoid producing scalding hot water.



**A mixing valve must be installed in the Everhot domestic water system to avoid scalding.**

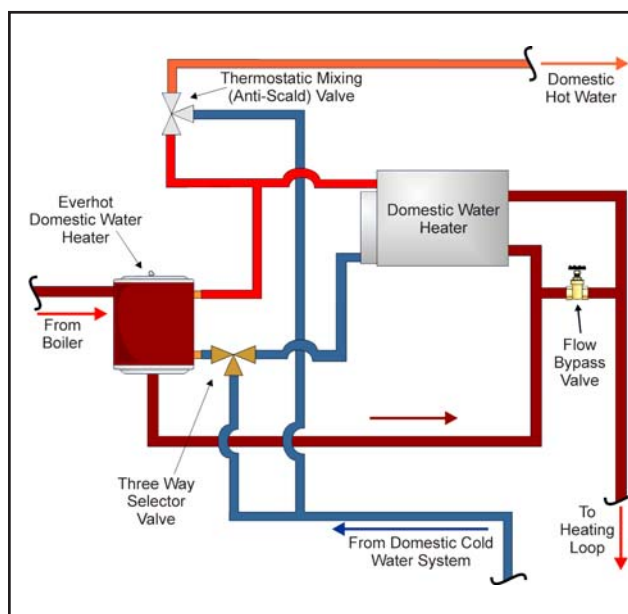


The Everhot has 1 1/4" boiler fittings for the water jacket and 3/4" or 1/2" fittings for the domestic water. Thanks to the full size boiler fittings and the low resistance within the water jacket an Everhot can be installed in the main water line or on a 3/4" loop. If installing in a loop, be sure that loop will be open at all times.

## Everhot/Domestic Hot Water Tank Combination

Both types of domestic water heat have their advantages. While an Everhot can provide an endless supply of hot water, it requires the operation of your boiler. A standard domestic water heater provides hot water independent of the heating system but it has a limited capacity of hot water and requires generator or shore power.

An Everhot and a standard domestic water heater can



be installed parallel to each other in the domestic water system. Although this system requires more space, the advantages of either system can be used with the turn of a valve.

A three-way valve is required to use as a selector between the systems. This valve should be installed in a readily accessible area and clearly labeled so you can tell which system is selected.



## Buffer Tank

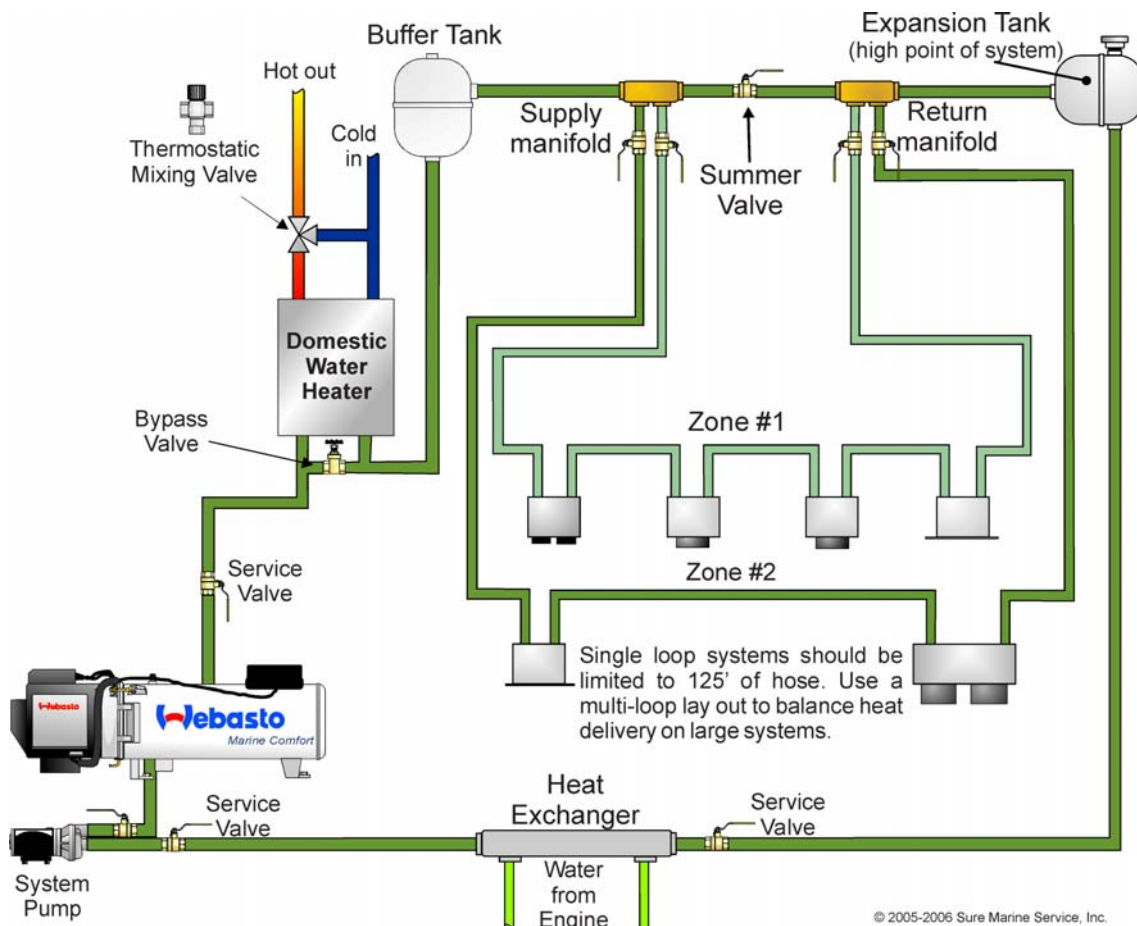
Common methods to increase water capacity of the heater system are installation of Everhot systems and/or the installation of a "buffer tank" that keeps heat in the system (Thermal Mass) and flowing throughout your boat without continual heater operation.

Though weight and space are considerations in your boat, SMS recommends that as much water as reasonably possible be used in your heating system. This will cut down on the possibility of your system "short cycling". Short cycling is a symptom of not enough water flow throughout the system and it can be very costly in repairs that may be incurred from this abbreviated run time. To avoid this problem, more is better in your water system.

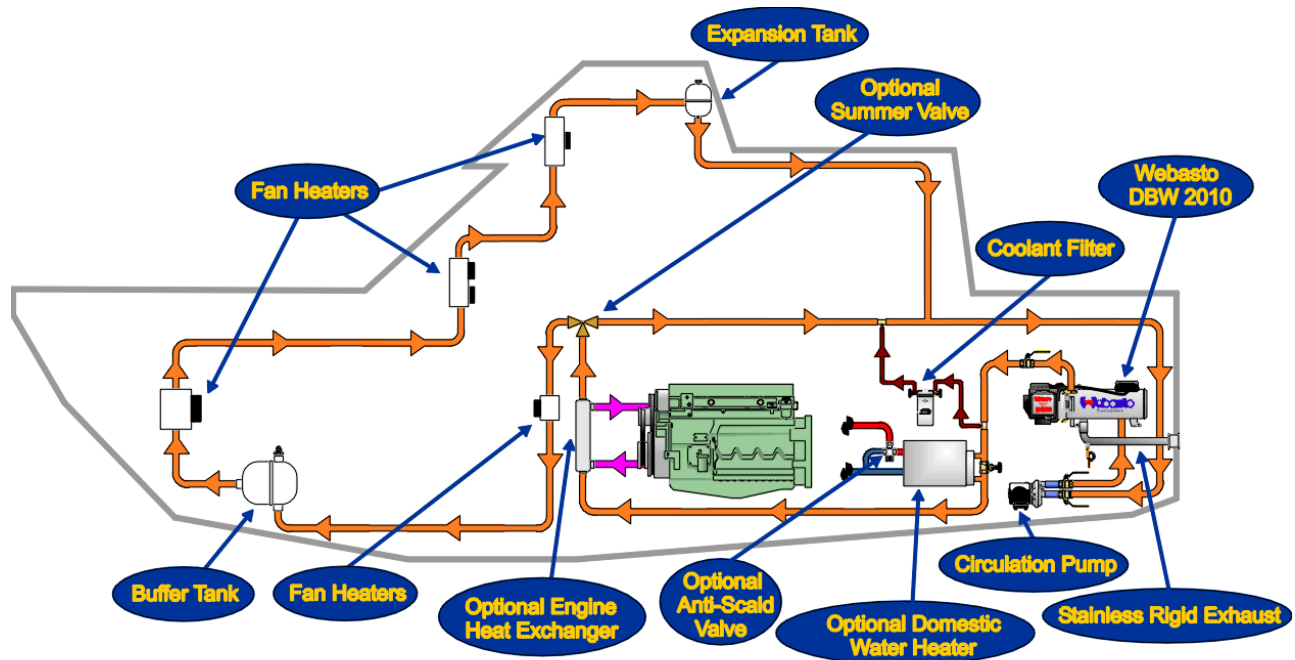


For the TSL-17, Thermo 90, DBW-2010 and DBW2015 we recommend a minimum of 10 gallons of water/antifreeze coolant.

We recommend that a DBW-2020 system hold at least 20 gallons of water/antifreeze coolant for proper operation, and the 300 have 30 gallons.



## Coolant Filter



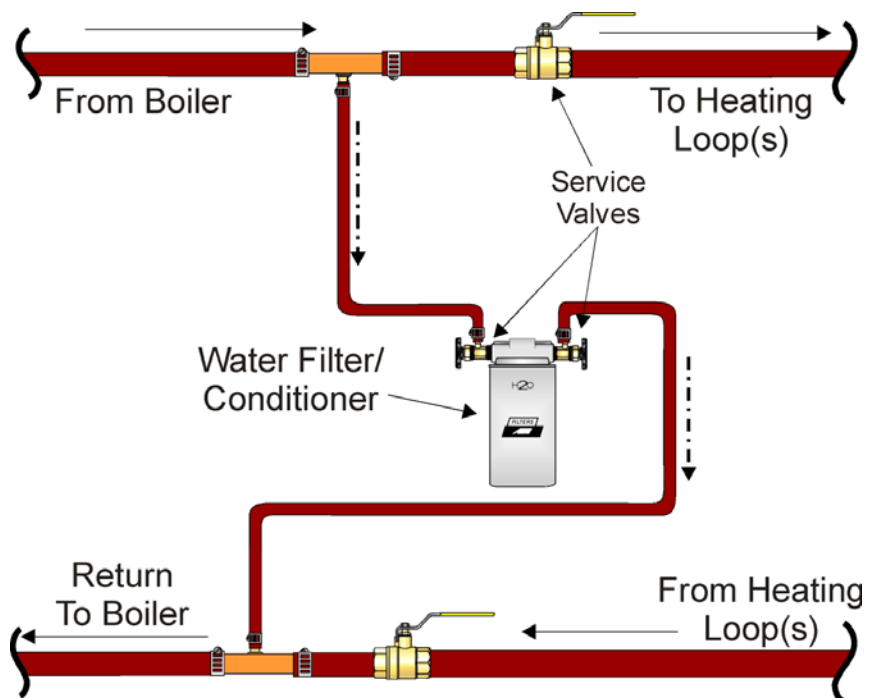
The coolant filter helps prevent corrosion by filtering and treating the coolant while the system is in operation. The filter canister contains an alkaline material that maintains a PH factor of 8 to 9. If the coolant becomes acidic (lower than 7 on the PH scale) corrosion of the heat exchanger within the water jacket can be expected.

The coolant filter kit comes with all proper parts for installation. The in line "T"s must be installed as noted in the diagram. The shut off valves should be installed on the filter assembly. They will help prevent losing coolant when changing the cartridge. The provided 3/8" hose should be cut in half for the installation.



**CAUTION:** Due to the size of the fittings and hose in the kit, the filter **MUST** only be installed in parallel as displayed and not in series. Installation in series will result in failure of your heating system.

The filter cartridge must be changed annually or as needed.



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## Water Filling Guidelines

The component listing below can help you figure the volume of water your heating system holds.

| COMPONENT            | CAPACITY             |
|----------------------|----------------------|
| Webasto Boiler       | 0.4 Qt.              |
| Everhot              | 5.7 gallons          |
| REAL Fan Heaters     | 0.2 quart ea.        |
| ¾" Heater Hose       | 2.97 gal per 100 ft. |
| ¾" Copper Tube       | 2.29 gal per 100 ft. |
| 1 ¼" Flex hose       | 6.4 gal per 100 ft.  |
| 5 gal Expansion tank | 5.2 gallons (full)   |

During normal operation the system should have a 60/40 water/antifreeze mix.

Multiply the capacity of your system by 0.4 to determine how much antifreeze you will need.

For example if you calculated that your system holds 19 and 1/2 gallons then you will need 7.8 gallons of antifreeze ( $19.5 \times 0.4 = 7.8$ ).

Have someone get 8 gallons of low silicate antifreeze but don't add it yet! Test for leaks FIRST.

## Fill - Check - Flush

The system should be flushed to expel any hose shavings, sawdust and so on that may have been introduced during installation.

First remove the expansion tank's cap to allow air to escape. Connect city water to the low point fill/drain valve and fill the system until water is to the ¼-full mark in the expansion tank.

Allow city water pressure to push water through the system.

## Do not energize your system pump.



Any foreign object in the system can foul or damage the impeller.

While the system is filling you should be checking for leaks. You want to find any leaks or kinks at this point instead of when the system has been filled with antifreeze. Once the inspection is completed, drain the water from the system.

## The water pump should never be run when the system is dry.



If you noticed any leaks or kinks you will want to do this again after you make your correction. Remember, at this point to use water only because you will be draining the system again.

## Bleeding the Water System

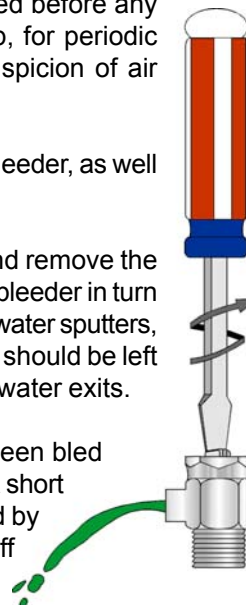
After flushing, use water to refill the system to the ¼ level in the expansion tank.

The water system should be bled before any operational tests, initial start up, for periodic maintenance and if there is suspicion of air trapped in the system.

Each fan heater should have a bleeder, as well as all high-point loops.

Make sure the system is cool and remove the expansion tank cap. Open each bleeder in turn and allow water to flow out. If the water sputters, there is air present. The bleeder should be left open until a constant stream of water exits.

To make sure the system has been bled properly, energize the pump for a short time. Trapped air may be moved by the water flow. Turn the pump off and operate each bleeder again to remove any trapped air.



If one loop refuses to bleed properly, close the other loops and energize the pump to force water through loop giving you trouble. Remember to turn the other loops on when bleeding is complete.

## Commissioning

### Water System Check

This test should be done after the water system has been flushed, all other systems have been properly installed, and the entire system is ready for start up.

Fill the system with water only and bleed, then run an operational test.

Test with water only in order to minimize damage or pollution if the system leaks when up to temperature, or if there is an unforeseen water system failure.

**Water alone should never be used as the operational coolant.**



Using only water will lead to corrosion in the system and will shorten the operational life of the components in the system.

With water only in the system, send power to the system and turn all thermostats up.

As the boiler begins to heat the system, inspect the entire installation. The boiler must have no oil, exhaust, or water leaks. Check all hose for kinks and all fittings for leaks.

Continue your inspection until the system is running at full operational temperature.

With an aquastat in the system, the fan heaters will not turn on until the coolant has warmed to 120°F.

When the system is fully operational there should be a 10°F difference between the coolant return and supply at the boiler. This difference is called the  $\Delta T$  (Delta T).

If the  $\Delta T$  exceeds 17°F, coolant flow is too low and the system will not function properly.

When you have completed your test turn all thermostats down and shut the system down to allow the water system to cool.

After the system has cooled down, check the water level in the expansion tank. If the system has lost a

substantial amount of water, there may be a leak in the system. After the leak has been located and fixed, another inspection should be conducted.

**Never check the system water level when the system is hot.**

Checking a hot system may result in scalding from escaping water or steam.

Once the system has been cleared for any leaks, antifreeze should be added.

The operating water system should be a 60% water 40% antifreeze mix. This is to avoid freezing of the system and ensure lubrication of the pump. The antifreeze should be a low-silicate antifreeze. Standard auto antifreeze contains high silicate counts that may leave deposits in the system and cause flow problems.

**The operating water system should never exceed 50/50 water to antifreeze mix.**



Using the total volume of the system figured before the test, drain 40% of the water in the system. For example, if you calculated that your system holds 21 and 1/4 gallons of water then you would want to drain out 8 and 1/2 gallons of water ( $21.25 \times 0.4 = 8.5$ ).

Close the drain and add antifreeze until you reach the 1/2 full level in the expansion tank. Leaving the expansion tank cap off, energize the system pump, but not the heater. Bleed the system of any air.

Once the system has been bled, the level in the expansion tank may have gone down. Add more antifreeze and bleed the system until the expansion tank is half full when the system is cool.

The coolant should be maintained at a PH factor of 8 to 9. If the coolant becomes acidic (lower than 7 on the PH scale) corrosion of the metal components within the system can be expected. Periodically the PH level of the coolant should be checked and adjusted with a coolant additive if below 8 on the PH scale.





## Post Installation

Notes:

Notes:



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